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Hyperactivated Glycolysis Drives Spatially-Patterned Kupffer Cell Depletion in MASLD

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eLife Assessment

The authors aim to understand why Kupffer cells (KCs) die in metabolic-associated steatotic liver disease (MASLD). This is a **valuable** study using in vitro studies and an in vivo genetic mouse model, suggesting that increased glycolysis contributes to KC death in MASLD. The data presented are now **convincing** and adequately revised. This work will be of interest to researchers in the immunology and metabolism fields.

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Abstract

Metabolic dysfunction-associated steatotic liver disease (MASLD) progression is characterized by hepatic inflammation and cell death, yet the mechanisms underlying Kupffer cells (KCs) loss remain poorly understood. Here, we sought to elucidate the metabolic basis of KCs death during MASLD. Using metabolomics, immunostaining, and flow cytometry, we evaluated metabolic alterations and KCs death throughout early MASLD progression. We found that KCs death is an early hallmark of MASLD, exhibiting greater susceptibility and displaying a spatial distribution consistent with KCs zonation. Moreover, KCs undergo progressive metabolic reprogramming toward enhanced glucose utilization during MASLD development, which is correlated with KCs death. In combination of biochemical agonist, isotope tracing and primary KCs culture, we further demonstrated that augmented glycolytic metabolism directly drives KCs death *in vitro*. Consistently, using *Chil1*-deficient mice, we further demonstrated that increased glucose utilization accelerates KCs death *in vivo*. Together, these findings establish a causal link between glycolytic activation and KCs loss during MASLD progression, highlighting glucose metabolic pathways as potential therapeutic targets to preserve KCs homeostasis and mitigate MASLD.

Introduction

Metabolic dysfunction-associated steatotic liver disease (MASLD) emerges as a prevalent liver condition in the western world, affecting roughly one-third of the population^{1–3}. Its incidence continues to climb due to its strong association with obesity, type 2 diabetes, and the metabolic syndrome^{4–6}. MASLD includes a spectrum of liver disorders, from metabolic dysfunction-associated fatty liver (MAFL), often referred to as steatosis, to metabolic dysfunction-associated steatohepatitis (MASH)⁷. MASH, characterized by steatosis, inflammation, ballooning injury, and varying degrees of fibrosis, marks the initial critical stage of MASLD⁸. Kupffer cells (KCs) are liver-resident macrophages located in the hepatic sinusoids^{9,10}. Derived from the embryonic yolk sac, KCs can self-renew through proliferation during adult homeostasis¹¹. Studies suggest that KCs contribute to triglyceride storage during MASH progression, as demonstrated by depleting CD207⁺ KCs in CD207-DTR mice using diphtheria toxin (DT) or by using CD207^{ΔBcl2l1} mice to stimulate a low embryo-derived KCs (EmKCs) status¹². Upon KCs death, monocyte-derived macrophages (MoMFs) gradually seed the KCs pool and eventually replace the deceased KCs¹³. These MoMFs tend to be more inflammatory than EmKCs, altering the liver's response in MASH, eventually

limiting triglyceride storage and contributing to liver fibrosis^{12,14}. Researchers, including our team, have observed a gradual decline in KCs during MASLD development^{12,14–17}. However, little is known regarding the dynamic loss of KCs and metabolic changes behind KCs death during MASLD.

Emerging evidence highlights a fundamental role of glucose metabolism in regulating macrophage function, polarization, and survival^{18–21}. Metabolic reprogramming is a well-established hallmark of macrophage activation and functional adaptation across diverse contexts²². Within the unique metabolic milieu of the MASLD liver, characterized by lipotoxicity, insulin resistance, and disrupted nutrient fluxes, it is plausible that KCs metabolism is profoundly altered. However, whether these metabolic perturbations directly contribute to the observed loss of KCs, and through which specific metabolic pathways, remains unclear. Therefore, this study aims to investigate the role of glucose metabolism in governing KCs susceptibility to death during MASLD progression. We sought to define the glucose metabolic alterations that occur in KCs as MASLD develops and to establish a causal link between metabolic reprogramming and KCs loss. Elucidating these mechanisms is essential for advancing our understanding of MASLD pathogenesis and for identifying potential therapeutic targets to preserve KCs homeostasis and mitigate disease progression.

Materials and methods

Animal experiments and procedures

Animals *Chil1*^{−/−} (strain no. T014402), *Chil1*^{flox/flox} (strain no. T013652), and *Clec4f-cre* [C-type lectin domain family 4-Cre (*Clec4f-cre*), strain no. T036801] mice with a *C57BL/6J* background were purchased from GemPharmatech. Accordingly, *C57BL/6J* mice (strain no. N000013) were used as wild-type (WT) mice. To generate *Clec4f^{ΔChil1}* mice, *Chil1*^{flox/flox} mice were crossed with *Clec4f-cre* mice and knock out efficiency was examined in KCs previously¹⁷. All mouse colonies were maintained at the Animal Core Facility of Yunnan University. The animal studies were approved by the Yunnan University Institutional Animal Care and Use Committee (IACUC, Approval No. YNU20220314). Male mice aged 6–8 weeks were used in this study. For 2-Deoxy-D-glucose (2-DG) Treatment in mice, WT *C57BL/6J* mice were fed a HFHC diet for five weeks. Beginning at week three, mice received intraperitoneal injections of either vehicle or 2-DG (50 mg/kg) every other day for two weeks. Mice were then euthanized for analysis of KCs viability and liver pathology.

Construction of MASLD/MASH mouse model Mice were provided a high-fat and high-cholesterol diet (HFHC, Research Diet, d12108c, 40 kcal% fat and 1.25% cholesterol) or a high-fat diet (HFD, Research Diet, d12492, 60 kcal% fat). Another group of mice was fed a methionine and choline deficient diet (MCD, Research Diet, A02082002BR). Throughout the feeding period, the body weight and food consumption of the mice were observed and recorded weekly. Once the dietary intervention was completed, the mice were euthanized. Liver and murine serum samples were collected for further analysis. Alanine aminotransferase (ALT) and aspartate aminotransferase (AST) levels in the serum, as well as cholesterol (TC) and triglyceride (TG) levels in both serum and liver tissues, were quantified using commercially available kits (Nanjing Jiancheng Bioengineering Institute). Genotyping Sample preparation and procedure were conducted as previously described¹⁷.

Data Presentation and Statistical Analysis

Data in graph figures are presented as mean ± standard error of the mean (SEM). Statistical analyses were performed using SPSS Statistics (Version 22). For comparisons between two groups, an unpaired two-tailed Student's t-test was used, while one-way analysis of variance (ANOVA) was applied for comparisons involving three or more groups. A p-value < 0.05 was considered statistically significant, with p-values indicated where applicable. All cell culture experiments were repeated at least three times independently. Figure 7  was created using the Figdraw platform (www.figdraw.com .

Additional Methods

Additional detailed methods can be found in the Supporting Information.

Results

The death of Kupffer cells is a pathological characteristic during MASLD

To systematically investigate KCs death, we established an MASLD mouse model by feeding wildtype C57Bl/6J mice a high-fat high-cholesterol diet (HFHC) (Figure S1A). We then performed Hematoxylin and eosin (H&E), Oil Red O and Sirius red staining to assess immune cell infiltration, fat accumulation, and liver fibrosis in mice fed HFHC for 0, 4 or 16 weeks. After 4 weeks of HFHC feeding, we observed a slight increase in lipid droplets, with no apparent signs of liver inflammation or fibrosis in hepatocytes (Figure S1A). By 16 weeks of HFHC feeding, both lipid droplets and immune cell infiltration had increased significantly, though liver fibrosis — as indicated by Sirius red staining, which labels collagen deposition — was still not induced, and MASLD activity score remained below 4 (Figure S1A). Moreover, the body weight of mice fed HFHC gradually increased over the feeding period (Figure S1B). Analysis of serum alanine aminotransferase (ALT), aspartate aminotransferase (AST), serum and liver cholesterol or liver triglyceride (TG) levels revealed significant increases compared to mice fed a normal chow diet (NCD), except for serum TG, which was similar between NCD and HFHC-fed mice (Figure S1C). These findings collectively indicate the successful establishment of an early MASLD mouse model, without fibrosis development.

Subsequently, we investigated KCs death by labeling KCs with antibodies targeting T cell immunoglobulin mucin protein 4 (TIM4)^{12,23}, and employing TdT-mediated dUTP Nick-End Labeling (TUNEL) to detect dead cells. Compared to baseline (0 weeks, prior to HFHC diet), KCs are significantly reduced with MASLD progression. Approximately 25% and nearly 60% of KCs underwent cell death by 4 and 16 weeks post-HFHC feeding, respectively (Figure 1A, 1B). To further validate KCs death during MASLD progression, we performed co-staining of the apoptotic marker cleaved caspase-3 (Cl-Casp3) with C-type lectin domain family 4 (Clec4f), another marker specific for KCs¹³. Although the ratio of Cl-Casp3⁺/Clec4f⁺ was relatively lower, its occurrence and gradual increase followed a trend similar to that of the TUNEL staining (Figure S2A). Considering the potential recruitment of MoMFs into the liver, some of which may acquire KCs features, including TIM4 and Clec4f expression^{12,14}, we performed flow cytometry analysis of liver nonparenchymal cells (NPCs) to distinguish resident KCs from monocyte-derived populations. Resident KCs were identified as CD45⁺ F4/80^{hi} CD11b^{low} TIM4^{hi} cells, MoKCs as CD45⁺ F4/80^{hi} CD11b^{low} TIM4^{low} cells, and infiltrating macrophages (IMs) as CD45⁺ F4/80⁺ CD11b^{hi} TIM4⁺ cells (Figure 1C). Our analysis showed that all TIM4⁺ KCs were of EmKCs, with no detectable MoKCs at any time point examined (Figure 1C). Consistent with the immunostaining results, the number of resident KCs decreased as early as 4 weeks and continued to decline through 16 weeks of HFHC feeding (Figure 1C, 1D). In contrast, IMs progressively accumulated during HFHC feeding (Figure 1C, 1D). To further refine the identification of MoMFs, we exclude CD45⁺ F4/80⁺ CD11b^{hi} TIM4⁺ Ly6G⁺ neutrophils from the IMs gate. This refined analysis yielded similar results, confirming a gradual increase in MoMFs over the course of HFHC feeding (Figure S2B, S2C).

To exclude TUNEL false positivity associated with proliferation, we evaluated Ki67 expression in dying KCs (TUNEL⁺TIM4⁺). We observed that proliferating KCs (Ki67⁺TIM4⁺) were infrequent throughout HFHC feeding, and less than 15% of TUNEL⁺TIM4⁺ KCs co-expressed Ki67. This indicates that the majority of TUNEL⁺ KCs represent true cell death events rather than proliferation-related artifacts (Figure S2D). To examine the broader occurrence of this phenomenon in MASLD, we further assessed KCs death in several other dietary mouse models, including mice fed a high-fat diet (HFD) for 20 weeks²⁴ and a methionine/choline deficient diet (MCD) for 6 weeks¹². Co-staining of TIM4 and TUNEL in liver sections revealed a notable increase

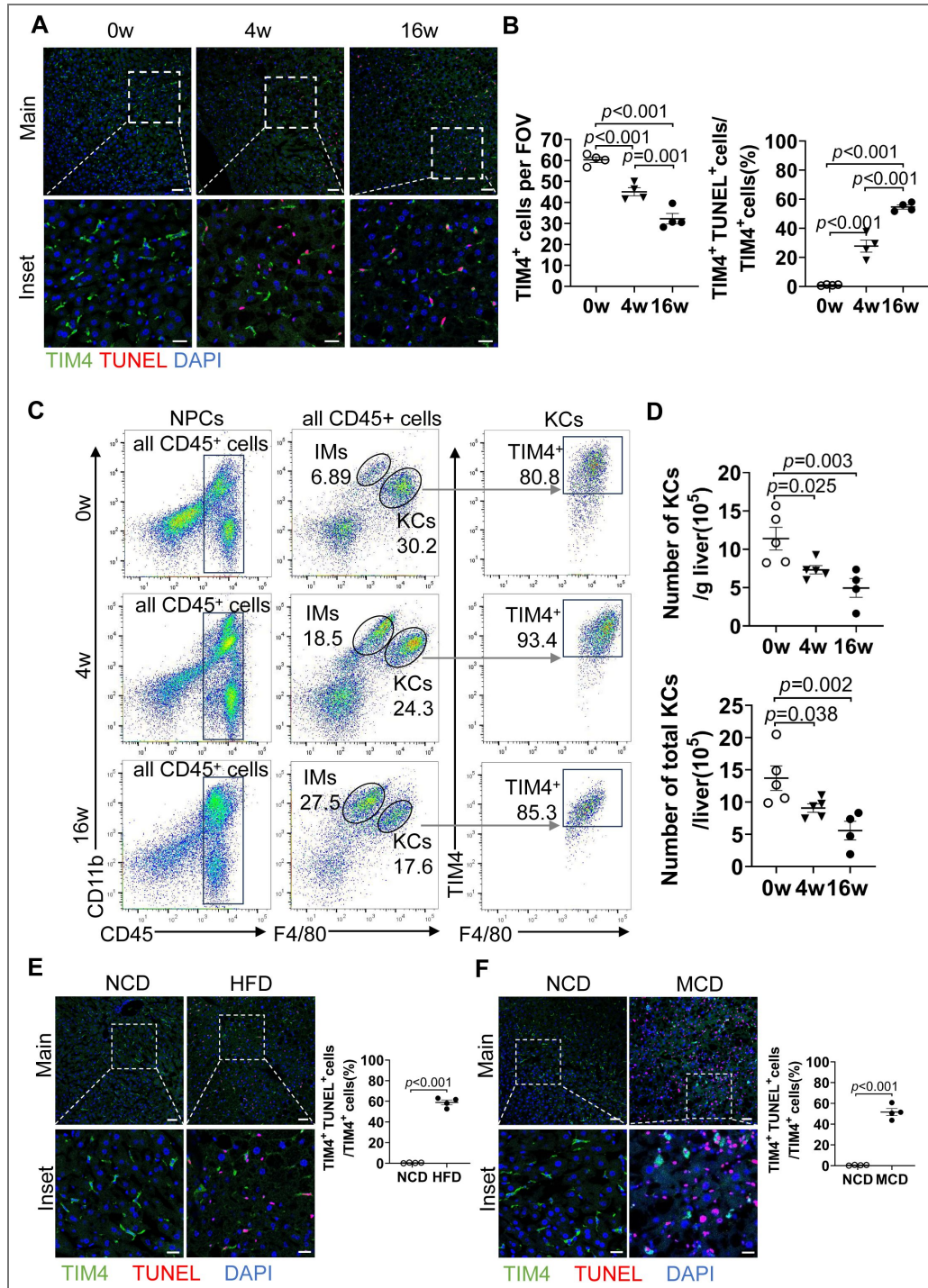


Figure 1. Kupffer cell death is a characteristic feature of MASLD progression.

(A–D) Male Wild-type C57BL/6j mice were fed a high-fat high-cholesterol diet (HFHC) for 0, 4, or 16 weeks. (A) KCs death was assessed by immunostaining of liver sections for TIM4 (KCs marker, red), TUNEL (green), and DAPI (nuclei, blue). Scale bar: 50 μ m (main panels) and 20 μ m (Inset). (B) KCs death was quantified. $n=4$ mice/group. (C) Flow cytometry analysis of KCs (CD45⁺ F4/80^{hi} CD11b^{low} TIM4⁺) and IMs (CD45⁺ F4/80^{low} CD11b^{hi} TIM4⁺) among isolated NPCs. (D) KCs counts were quantified. $n=4-5$ mice/group. (E–F) Male wild-type C57BL/6j mice were fed either: (E) Normal chow diet (NCD) or high-fat diet (HFD) for 20 weeks, or (F) NCD or methionine-choline-deficient diet (MCD) for 6 weeks. KCs death was assessed by immunostaining of liver sections for TIM4 (green), TUNEL (red), and DAPI (nuclei, blue). Scale bar: 50 μ m (main panels) and 20 μ m (Inset). KCs death was quantified. $n=4$ mice/group. Representative images are shown in A, C, E, F. One-way ANOVA (B, D). Unpaired Student's t-test (E, F). P value as indicated.

in KCs death in both the HFD and MCD groups compared to the control group fed a normal chow diet (NCD) (Figure 1E, 1F). These findings collectively confirm that progressive KCs death is a pathological hallmark of MASLD observed across various dietary-induced models.

Kupffer cells exhibit early and zone-specific susceptibility to death during MASLD progression

To compare relative susceptibility of different hepatic cell populations to MASLD-induced death, we performed TUNEL co-staining with lineage markers: HNF4 α (hepatocytes), Desmin (hepatic stellate cells; HSCs), and Iba1 (Hepatic macrophages, including both KCs and MoMFs) (Figure 2A–C). Hepatocyte death showed minimal change during the initial 4 weeks but increased modestly by 16 weeks (Figure 2A). In contrast, HSCs and hepatic macrophages mortality rose progressively throughout MASLD progression, with significant increases detectable as early as 4 weeks (Figure 2B, 2C). Notably, the finding that 60% of TIM4⁺ KCs are TUNEL⁺, compared to 30% of total Iba1⁺ cells, further supports that resident KCs undergo death more readily than the broader macrophage pool, which includes MoMFs.

We next determined whether KCs death displays spatial zonation within the liver lobule. Based on the lobular architecture—comprising periportal (PP, zone 1), midzonal (Mid, zone 2), and pericentral (PC, zone 3) regions—we performed co-staining for TIM4, TUNEL, and glutamine synthetase (GS; a central vein marker) (Figure 2D). Spatial analysis revealed an early zonation bias in KCs death. At 8 weeks of HFHC feeding, the proportion of TIM4⁺TUNEL⁺ cells among total TIM4⁺KCs was significantly higher in PV regions compared to CV regions ($p = 0.041$), indicating increased susceptibility of periportal KCs (Figure 2D). However, by 16 weeks, although absolute numbers of apoptotic KCs differed across zones, the proportional death rate became comparable (Figure 2D), suggesting that prolonged HFHC feeding leads to widespread KCs loss that attenuates early zonation bias.

Kupffer cells exhibit metabolic reprogramming with increased glycolysis during early MASLD

To define the glucose metabolic alterations that occur in KCs during MASLD development, we isolated KCs from wildtype mice at various time points of HFHC feeding (0, 4, 8, and 16 weeks). The purity of isolated KCs was confirmed by TIM4 immunofluorescence staining, with TIM4⁺ cells exceeding 90% (Figure S3A). We then conducted qRT-PCR to assess the mRNA expression levels of key enzymes involved in glycolysis (*Slc2a1*, *Hk3*, *Pfkfb3*, *Pkm*), the pentose phosphate pathway (PPP) (*G6pd*, *6pdg*), glycogenolysis (*Pygl*), and glycogenesis (*Gys1*, *Ugp2*). Our data revealed that mRNA expression of rate-limiting enzymes for fast glucose metabolism, such as glycolysis and PPP, was significantly increased as early as 8 weeks after initiating the HFHC diet (Figure S3B). While glycolytic enzyme expression remained elevated, PPP enzyme expression began to decline by 16 weeks (Figure S3B). In contrast, enzymes linked to slower glucose metabolism, such as oxidative phosphorylation (*Idh1*, *Ogdh*), did not exhibit significant changes (Figure S3B). Furthermore, mRNA expression of glycogenesis and glycogenolysis rate-limiting enzymes started to decrease at 8 and 16 weeks, respectively, suggesting that glucose uptake becomes the primary source of KCs glucose metabolism during this period (Figure S3B). Additionally, we investigated mRNA expression of rate-limiting enzymes involved in β -oxidation (*Acadm*, *Hadh*) but found no significant differences (Figure S3B). These data suggest a time-dependent metabolic inflexibility, where impaired glycogen handling and mitochondrial inertia drive a sustained glycolytic dependence in KCs, which may exacerbate their vulnerability (particularly in periportal zones, where glycogen storage is predominant²⁶).

To validate KCs-specific metabolic alterations in MASLD, we performed metabolomic analysis on primary KCs from wild-type mice fed an HFHC diet for 0, 4, or 8 weeks (Figure 3A). Given that KCs death peaked at 8 weeks in our model (Figure 2D), we focused on these early time points to capture initial metabolic shifts. Principal component analysis (PCA) of KCs metabolites revealed distinct, diet duration-dependent clustering (Figure 3B). Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis identified glucose metabolism pathways—including glycolysis and the pentose phosphate pathway (PPP)—as the most significantly

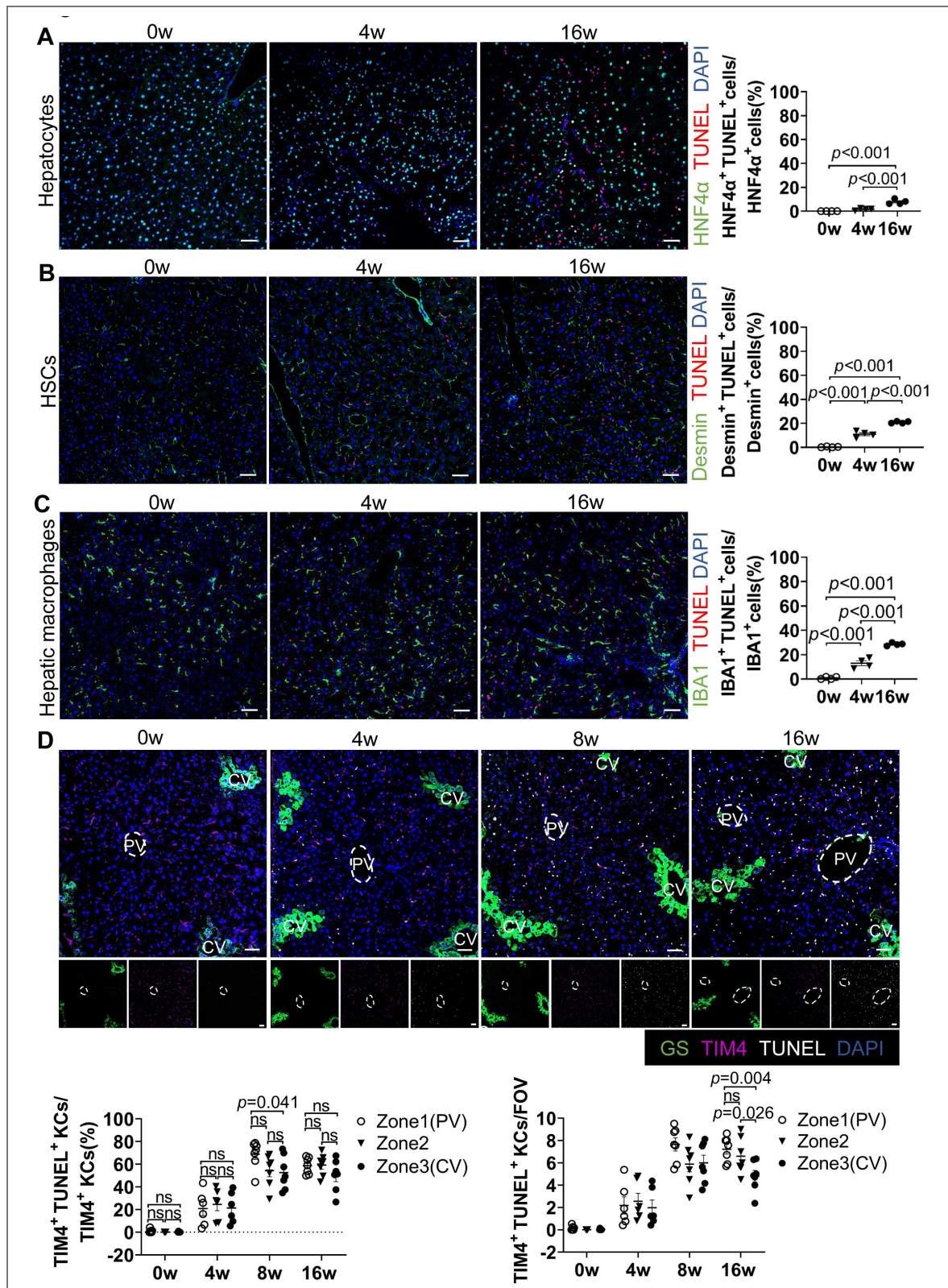


Figure 2. Kupffer cells exhibit early and zone-specific susceptibility to death during MASLD progression.

(A-E) Male Wild-type C57BL/6J mice were fed a HFHC diet for 0, 4, or 16 weeks. (A-C) Hepatic cell death was assessed by co-staining TUNEL with: (A) HNF4α (hepatocytes), (B) Desmin (hepatic stellate cells, HSCs), (C) Iba1 (hepatic macrophages), and DAPI (nuclei, blue). Scale bars: 50 μm (main panels). Hepatic cell death was quantified (n = 4 mice/group). (D) Zonal distribution of KCs death was evaluated by co-staining TIM4 (KCs), TUNEL, Glutamine Synthetase (GS, central vein marker) and DAPI (nuclei, blue). Scale bars: 50 μm. Zonal distribution of KCs death was quantified (n = 6-7 mice/group). FOV: field of view. PV: portal vein. CV: central vein. Representative images are shown in A-D. One-way ANOVA (A-D). P value as indicated.

upregulated during MASLD progression (Figure 3C, 3D). This glycolytic activation was further corroborated by time-dependent accumulation of key intermediates including glucose, Phosphatidylethylamine [PEA], Phosphoenolpyruvate [PEP], fructose-1,6-bisphosphate [FBP], and lactate [LA] in heatmap analysis (Figure 3E). Critically, we observed progressive increases in pro-apoptotic metabolites generated through these glucose metabolism pathways: redox disruptors (GSSG, FAD), mitochondrial toxins (methylmalonic acid), and apoptosis mediators (Hcy) - all exhibiting temporal coupling with glycolytic intermediates (Figure 3F). This demonstrates that KCs undergo rapid glycolytic reprogramming during early MASLD pathogenesis, which actively generates cytotoxic effectors coinciding with their peak vulnerability.

Excessive glucose metabolic activity contributes to Kupffer cells death

To investigate the direct role of glucose metabolism in KCs death, isolated KCs were subjected to *in vitro* metabolic perturbations. First, KCs were treated with a combination of high glucose and palmitic acid (PA) to model MASLD pathology. Physiological glucose (5.5 mM) combined with PA (800 μ M) increased KCs death by approximately 10%. Strikingly, glucose at a concentration mimicking HFHC feeding (10 mM) combined with PA (800 μ M) increased KCs death by approximately 27%. This was evidenced by increased Cl-Casp3 staining (Figure 4A) and elevated Cl-Casp3 protein levels via western blot (Figure 4B).

Second, we treated KCs with the Pyruvate Dehydrogenase Kinase (PDK1) activator PS48, which directly stimulates glycolysis. The results indicated a markedly increased KCs death compared to blank or DMSO vehicle controls (Figure 4C, 4D). Finally, Oligomycin (an ATP synthase inhibitor) was used to force glycolytic reliance by blocking mitochondrial ATP production. Oligomycin treatment similarly induced significant KCs death (Figure S4A, S4B), phenocopying the effects of direct glycolytic activation. KCs are difficult to culture and prone to death *in vitro*. To assess the health status of KCs in our metabolic perturbation assays, we also examined KCs viability using Calcein-AM staining, a fluorescent dye that labels metabolically active cells. Image analysis confirmed that the KCs were healthy and alive under all tested conditions (Figure S4C), indicating that our isolated KCs were suitable for *in vitro* experiments.

To determine whether enhanced glycolytic activity contributes to KCs death *in vivo* during MASLD, we pharmacologically inhibited glycolysis using 2-deoxy-D-glucose (2-DG). 2-DG functions as a glucose analog that enters cells but cannot be fully metabolized, thereby inhibiting glycolysis through blockade of hexokinase and phosphoglucose isomerase (Figure 4E). Wild-type C57BL/6J mice were fed a HFHC diet for five weeks to induce early-stage MASLD. To inhibit KCs glycolysis *in vivo*, mice received intraperitoneal injections of either vehicle or 2-DG (50 mg/kg) every other day beginning at week 3 of HFHC feeding. The relatively short duration of HFHC exposure (5 weeks) combined with low-dose 2-DG treatment was designed to minimize potential confounding effects of systemic glycolytic inhibition on hepatocytes. At the end of the five-week feeding period, KCs survival was assessed by co-immunostaining liver sections for the KCs marker TIM4 and TUNEL to detect apoptotic cells (Figure 4F). Quantitative analysis demonstrated that 2-DG-treated mice exhibited a significantly lower proportion of TUNEL⁺ KCs compared with vehicle-treated controls (Figure 4G). These results indicate that pharmacological inhibition of glycolysis protects KCs from cell death during MASLD. Together with our *in vitro* data showing that glycolytic activation promotes KCs death.

Enhanced glycolytic flux in *Chil1*^{-/-} macrophages

To investigate whether hyperactivated glycolysis specifically in KCs drives their death *in vivo*, we focused on Chitinase 3-like 1 (Chi3l1; gene *Chil1*)—a known inhibitor of glucose uptake¹⁷. Since Chi3l1 suppresses glucose uptake by KCs, KCs in *Chil1*^{-/-} mice are expected to maintain a state of hyperactivated glycolysis. To confirm this, we first performed a glucose metabolic flux assay in KCs. However, due to the limited availability of primary KCs, we used bone marrow-derived macrophages (BMDMs) to dissect the Chi3l1-dependent metabolic mechanisms. Uniformly labeled [U-¹³C]glucose tracer analysis (Figure 5A) revealed genotype-specific metabolic reprogramming: PCA showed distinct clustering between *Chil1*^{-/-} and wild-type (WT) BMDMs (Figure 5B), and heatmaps demonstrated pronounced accumulation of glycolytic intermediates (Figure 5C).

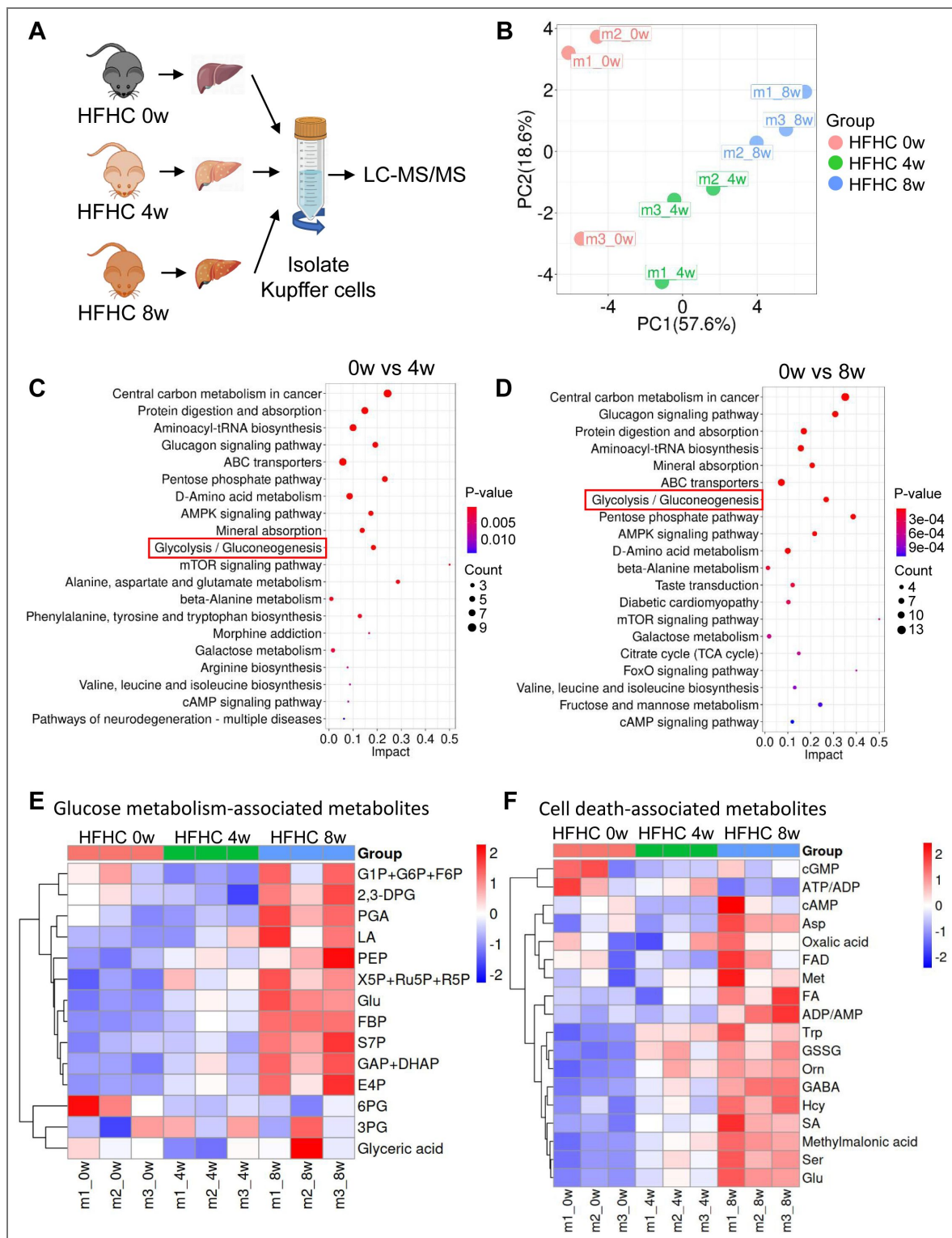


Figure 3. Kupffer cells exhibit metabolic reprogramming with increased glycolysis during early MASLD.

(A) Experimental design for metabolomic analysis of KCs isolated from male wild-type mice fed a HFHC diet for 0, 4 or 8 weeks. $n=3$ mice/group. (B) Principal component analysis (PCA) of enriched metabolites in KCs across different dietary durations. (C-D) Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis of metabolic pathways upregulated in KCs at 4 weeks (C) or 8 weeks (D). The glucose metabolism pathway is highlighted by red rectangles. (E) Heatmap depicting significantly altered metabolites involved in glucose metabolism pathways in KCs across different dietary durations. (F) Heatmap depicting significantly altered metabolites involved in cell death in KCs across different dietary durations.

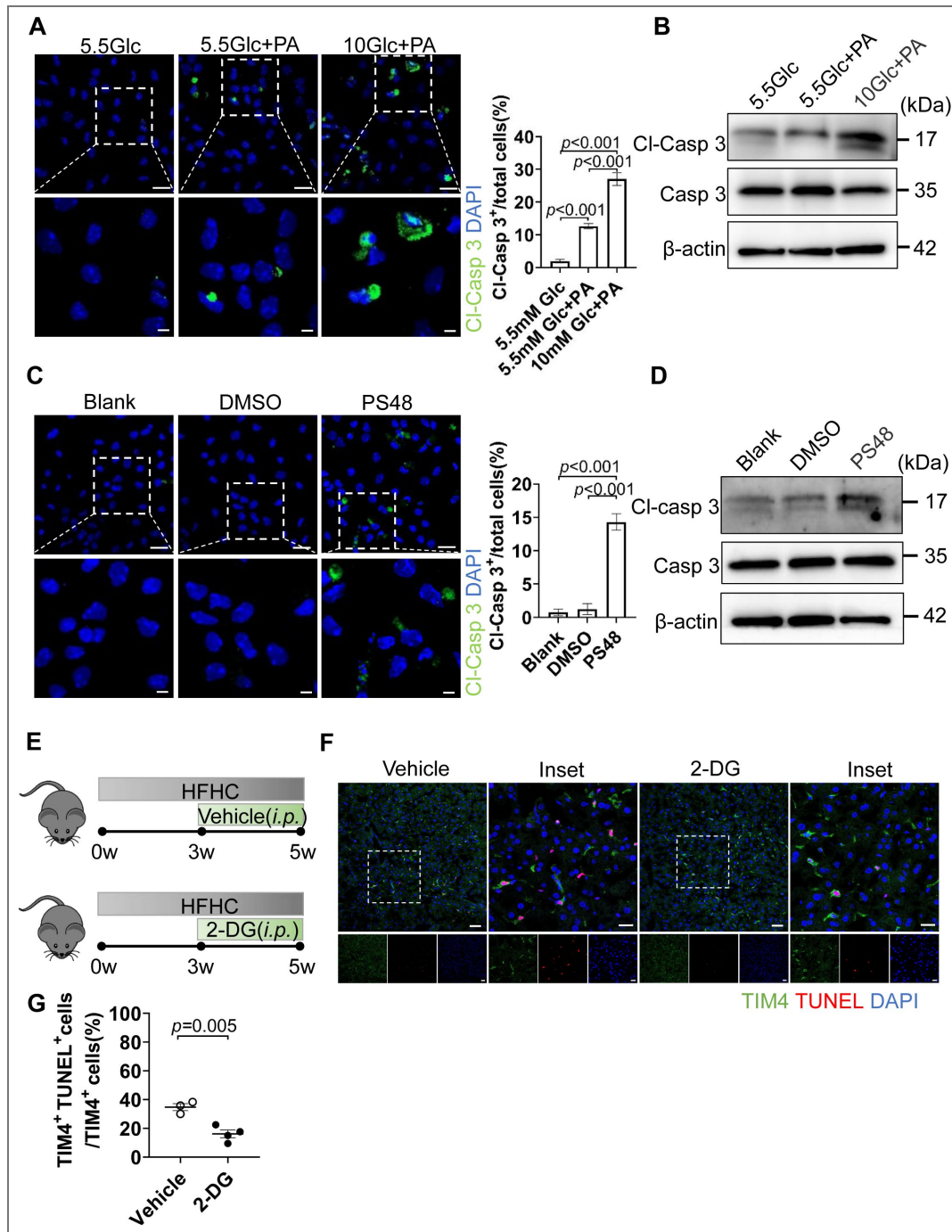


Figure 4. Excessive glucose metabolic activity contributes to Kupffer cell death.

(A–B) Isolated KCs were treated for 24 h with: 5.5 mM glucose + isopropanol (control), 5.5 mM glucose + 800 μ M palmitic acid (PA), 10 mM glucose + 800 μ M PA. Cell viability was assessed by Cleaved caspase-3 (CI-Casp3) staining (CI-Casp3⁺ cells = dead). Scale bars: 20 μ m (main panels), 5 μ m (insets). CI-Casp3 was detected by Western blot. (C–D) Isolated Kupffer cells were treated for 24 h with: Blank (no treatment), DMSO (vehicle control), 20 μ M PS48 (PDK1 activator). Scale bars: 20 μ m (main panels), 5 μ m (insets). Cell death were analyzed as above. (E) Experimental design. Male WT mice were fed a high-fat, high-cholesterol (HFHC) diet for 5 weeks. From the 3rd week onward, mice received intraperitoneal injections of either vehicle or 2-DG (50 mg/kg) every other day ($n = 3$ –4 mice/group). (F–G) Effects of glycolysis inhibition on KCs death after 5 weeks of HFHC feeding. (F) Representative images of liver sections co-stained with TUNEL and the KCs marker TIM4. (G) Quantification of TUNEL+ KCs. Data are presented as mean $\pm$ SEM. Statistical analysis was performed using one-way ANOVA (A, C) and an unpaired Student's *t*-test (G); *P* values are indicated.

Metabolic flux quantification confirmed significantly elevated U-¹³C enrichment in glycolytic metabolites—including glucose (Glc), fructose-6-phosphate (F6P), 3-phosphoglycerate (3PGA), 2-phosphoglycerate (2PGA), phosphoenolpyruvate (PEP), pyruvate (PA), and lactate (LA)—whereas glucose-6-phosphate (G6P), fructose-1,6-bisphosphate (FBP), and pentose phosphate pathway (PPP) intermediates (ribulose-5-phosphate [Ru5P], ribose-5-phosphate [R5P], sedoheptulose-7-phosphate [S7P]) remained unchanged. Dihydroxyacetone phosphate (DHAP) was the only PPP-linked metabolite showing increased enrichment (Figure 5D [↗](#)), indicating that *Chil1* deletion selectively hyperactivates glycolysis without engaging the PPP. Furthermore, extracellular acidification rates (ECAR) analysis also showed significantly elevated glycolytic capacity in *Chil1*^{−/−} BMDMs (Figure 5E, 5F [↗](#)).

Increased glycolysis is often associated with proinflammatory polarization of macrophages¹⁵. To assess the effect of *Chil1* loss on KCs polarization, we further isolated KCs from WT and *Chil1*^{−/−} mice fed a HFHC diet at 0, 8 and 16 weeks, and measured the mRNA expression of proinflammatory markers (*Nos2*, *Cxcl9*, *CIITA*, *CD86*, *Ccl3*, and *Ccl5*) and anti-inflammatory markers (*Chil3*, *Retnla*, *Arg1*, and *Mrc1*). In WT KCs, the mRNA levels of proinflammatory markers increased with MASLD progression: *Nos2*, *Cxcl9*, *CIITA*, and *Ccl3* rose significantly, while *CD86*, *Ccl5* showed an upward trend (Figure S5A [↗](#)). In contrast, *Chil1*^{−/−} KCs exhibited higher baseline expression of proinflammatory genes, with *Nos2* and *Ccl5* significantly elevated and *Cxcl9* and *CD86* showing an increasing tendency compared to WT. As MASLD developed, proinflammatory gene expression increased further in *Chil1*^{−/−} KCs. No significant differences in anti-inflammatory marker expression were observed between WT and *Chil1*^{−/−} KCs at baseline. However, with MASLD progression, *Chil1*^{−/−} KCs displayed a rising trend in the expression of anti-inflammatory markers, including *Chil3*, *Arg1*, and *Mrc1* (Figure S5A [↗](#)). Together, these data suggest that *Chil1* deficiency drives KCs toward a partially pro-inflammatory phenotype.

Since *Chil1* primarily functions in its secretory form as a macrophage glucose uptake inhibitor¹⁷, we also added recombinant *Chil1* (r*Chil1*) supplementation group. r*Chil1* supplementation reversed the hyper-glycolytic flux, restoring it to levels comparable to those in WT macrophages (Figure S6A–S6D [↗](#)). This was accompanied by reduced mass isotopologue distributions of key glycolytic intermediates (Glc, F6P, FBP, 3PGA, 2PGA, PEP, PA, LA, G6P) (Figure S6B, S6C [↗](#)). In contrast, PPP intermediates (Ru5P, R5P, S7P, DHAP) were unaffected (Figure S6B, S6C [↗](#)), confirming the specificity of the effect on glycolysis. Functionally, r*Chil1* significantly lowered lactate dehydrogenase (LDH) activity in high glucose-treated BMDMs, further corroborating the attenuation of glycolytic output (Figure S6D [↗](#)). We acknowledge that detailed metabolic flux analysis was performed in BMDMs rather than primary KCs due to the technical challenge of obtaining sufficient viable cells for such assays. While ontogenically distinct from KCs, BMDMs are a widely accepted model for mechanistic metabolic dissection. Importantly, the hyperglycolytic phenotype observed in *Chil1*^{−/−} BMDMs was consistent with our *in vivo* KCs data: *Chil1* deficiency drove pro-inflammatory polarization. Thus, despite this limitation, the concordance between our *in vitro* mechanistic findings and *in vivo* phenotypic data provides compelling evidence that *Chil1*^{−/−} mice serve as a mechanistic model to investigate glycolysis-driven KCs death during MASLD. Collectively, these data establish that *Chil1* reduces glycolytic flux—without affecting PPP activity—in macrophages, thereby providing a mechanistic model to investigate glycolysis-driven KC death.

Enhanced glycolysis drives Kupffer cell death in MASLD

We therefore employed *Chil1*^{−/−} mice as a mechanistic model to investigate glycolysis-driven KCs death. First, we isolated primary KCs from WT and *Chil1*^{−/−} mice and treated them with PA or Iso, or left untreated (Blank). *Chil1*^{−/−} KCs exhibited significantly increased susceptibility to PA-induced cell death compared to WT controls *in vitro*, as indicated by elevated Cl-Casp3 immunostaining (Figure 6A, 6B [↗](#)). This result was further confirmed by increased LDH release from *Chil1*^{−/−} KCs (Figure 6C [↗](#)). To delineate the cellular source of *Chil1* in MASLD livers, we performed immunostaining on serially sectioned liver tissues from mice fed an HFHC diet for 16 weeks. Consecutive sections were independently probed for *Chil1* or lineage-specific markers (HNF4α, Desmin, Iba1), enabling cellular localization through morphological alignment across sequential

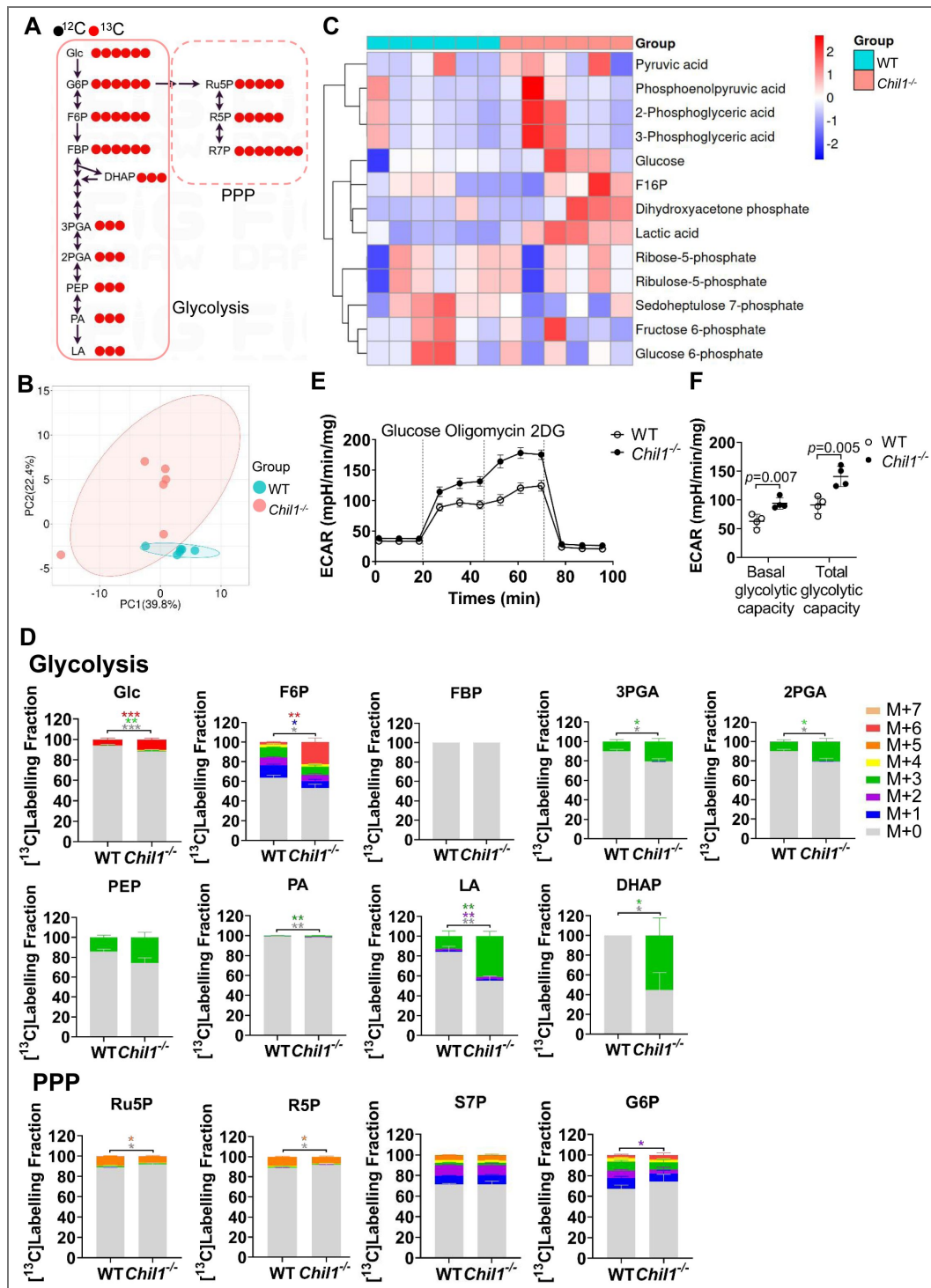


Figure 5. Enhanced glycolytic flux in *Chil1*^{-/-} macrophages.

(A) Schematic diagram depicting the fate of glucose-derived ribose carbons in WT mouse primary hepatocytes. (B) Principal component analysis (PCA) of metabolites in WT and *Chil1*^{-/-} BMDMs cultured with [U-¹³C]glucose. (C) Heatmap depicting significantly altered Glycolysis and Pentose phosphate (PPP) metabolites in WT and *Chil1*^{-/-} BMDMs. (D) Glucose metabolic flux analysis in WT and *Chil1*^{-/-} BMDMs cultured with [U-¹³C]glucose showing mass isotopologue distributions of: Glycolytic intermediates (Glc, F6P, FBP, 3PGA, 2PGA, PEP, PA, LA, G6P). PPP intermediates (Ru5P, R5P, S7P, DHAP). Data represent n = 6 biological replicates/group. (E-F) Extracellular acidification rate (ECAR) analysis of WT or *Chil1*^{-/-} BMDMs cells. BMDM were sequentially treated with Glucose, oligomycin and 2-DG as indicated during seahorse. Unpaired student t-test (D,F). P* < 0.05, P** < 0.01, P*** < 0.001.

slices. This revealed predominant Chi3l1 expression in hepatic macrophages (Figure S7A [↗](#)). Moreover, Chi3l1 expression was significantly elevated in KCs (F4/80⁺TIM4⁺) of HFHC-fed mice, compared to NCD controls (Figure S7B [↗](#)), suggesting Chi3l1 is primarily expressed in KCs and may function as a KCs-autonomous regulator.

To determine cell-autonomous effects, we generated KCs-specific *Chi1* knockout mice (*Clec4f*^{Δ*Chi1*} mice) and analyzed their response to HFHC feeding. Flow cytometry of NPCs revealed a significant lower KCs population (CD45⁺ F4/80^{hi} CD11b^{low} TIM4⁺) in *Clec4f*^{Δ*Chi1*} mice compared to *Clec4f*^{cre} controls after 16 weeks of HFHC diet (Figure 6D, 6E [↗](#)). During this analysis, we observed no reduction in KCs in *Clec4f*^{cre} control mice, raising the possibility that Cre insertion itself might influence KCs maintenance. To test this, we performed co-staining for TIM4 and Ki67, which revealed significantly higher numbers of proliferating KCs in *Clec4f*^{cre} mice compared with C57BL/6J mice under NCD. Notably, following HFHC feeding, *Clec4f*^{cre} mice exhibited an even greater increase in KCs proliferation (Figures S8A and S8B), suggesting that cre insertion enhanced KCs self-renewal which contributes to maintenance of the KCs pool in these mice under MASLD. Despite this proliferative baseline, *Clec4f*^{Δ*Chi1*} mice still displayed a reduced KCs number (Figure 6D, 6E [↗](#)) and increased KCs death. This was corroborated by TIM4/TUNEL co-staining, which showed elevated KCs death in *Clec4f*^{Δ*Chi1*} livers (Figure 6F, 6G [↗](#)). Collectively, these findings demonstrate that genetic ablation of *Chi1* amplifies glycolytic flux and sensitizes KCs to lipotoxic stress *in vitro*, and drives KCs depletion through accelerated cell death *in vivo*. Although Chi3l1 is not exclusively expressed by KCs, these data establish it as a critical regulator of KCs survival in MASLD, likely acting in a cell-autonomous manner within the KCs population.

Discussion

This study identifies KCs death as an early and selective pathological feature of MASLD across multiple dietary models. Compared with other hepatic cell populations, KCs exhibit markedly increased susceptibility to apoptosis during early disease. Through metabolomic and functional analyses, we demonstrate that KCs undergo progressive metabolic reprogramming characterized by sustained activation of glycolysis. Importantly, our data support a causal role for excessive glycolytic flux in driving KCs death. Pharmacologic glycolytic stimulation (high glucose or PDK1 activation with PS48), enforced glycolytic dependence (oligomycin), and KC-specific Chi3l1 deletion each accelerated KC apoptosis, whereas glycolytic inhibition (2-DG) was protective. Together, these findings establish hyperactivation of glycolysis as a central mechanism underlying KCs vulnerability and depletion in MASLD (Figure 7 [↗](#)).

KCs loss has been reported to varying degrees across MASLD models, reflecting differences in diet composition, disease duration, macrophage heterogeneity, and marker selection. For example, in a 6-week MCD diet model, approximately 10% of CLEC4F⁺ KCs were TUNEL⁺, indicating apoptotic death¹². In contrast, another 6-week MCD study reported a marked reduction in TIM4⁺ KCs, from 66% to 26%²⁷. More moderate KCs loss has been observed in HFD models, with a ~20% reduction of TIM4⁺ KCs after 16 weeks¹⁴, whereas prolonged western diet feeding resulted in a >50% decrease in TIM4⁺ KCs at 36 weeks¹⁴. Together, these studies underscore the model-dependent nature of KCs loss and highlight the importance of experimental context and marker selection when assessing KCs dynamics in MASLD. Importantly, our spatial analyses further reveal preferential periportal KCs death during early MASLD. Given the metabolic zonation of the liver and higher periportal glucose flux, this regional susceptibility supports a mechanistic link between glucose metabolism and KCs survival. These findings suggest that metabolic microenvironment, in addition to inflammatory signaling, shapes KCs fate during MASLD progression.

While glycolytic reprogramming is classically associated with macrophage activation¹⁵, our data reveal a critical distinction in tissue-resident KCs. In MASLD, sustained and excessive glycolytic activation does not simply maintain inflammatory function but instead drives apoptotic loss. Thus, glycolysis in this context becomes maladaptive, serving as a determinant of cell fate rather than merely an activation program. This shifts the conceptual framework from inflammation-centered KCs dysfunction to metabolism-driven KCs attrition. Our findings also clarify the context-dependent role of Chi3l1 in liver disease. Prior studies in advanced MASH and fibrotic models

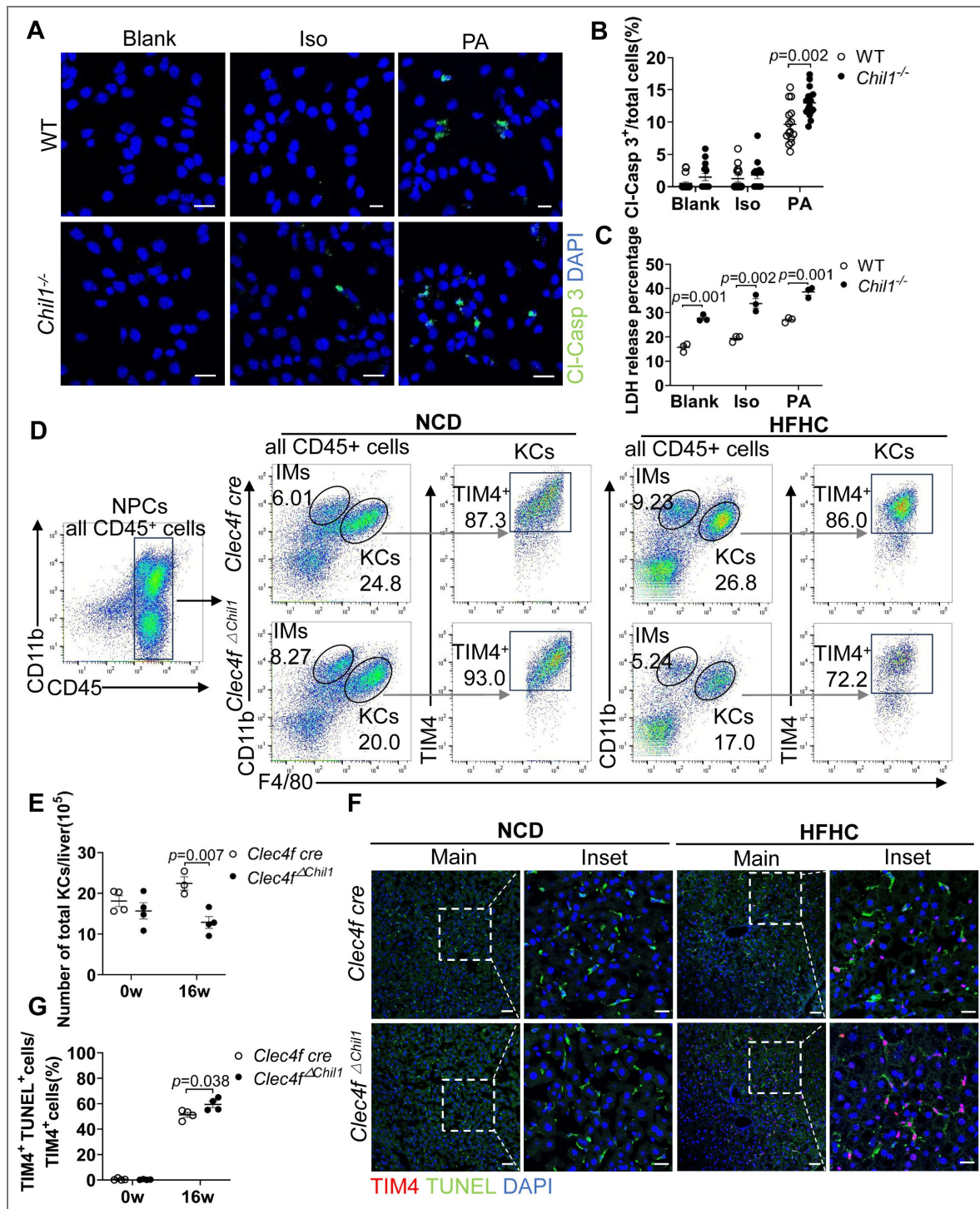


Figure 6. Enhanced glycolysis accelerated Kupffer cell death during MASLD.

(A) Cleaved caspase-3 (CI-Casp3) staining to detect WT and *Chil1*^{-/-} Kupffer cell death. Cells were under treatment without (Blank) or with either Isopropyl alcohol (Iso) or Palmitic Acid (PA) for 24 h. Scale bar: 20μm. (B) CI-Casp3⁺ cells were quantified. (C) LDH release measurement in culture medium of KCs isolated from male WT and *Chil1*^{-/-} mice was measured after treatment for 24 h with: Blank (no treatment), ISO (vehicle control), 800 μM PA. (D) Flow cytometry analysis of KCs (CD45⁺ F4/80^{hi} CD11b^{low} TIM4⁺) and MoMFs (CD45⁺ F4/80^{low} CD11b^{hi} TIM4⁻) among NPCs in *Clec4f*-cre and *Clec4f*^{Δ*Chil1*} mice fed HFHC diet for 0 or 16 weeks. (E) KCs counts were quantified. n=4 mice/group. (F) Kupffer cell death was assessed by immunostaining of TIM4 (KCs marker, green), TUNEL (red), and DAPI (nuclei, blue) in liver sections from *Clec4f*-cre and *Clec4f*^{Δ*Chil1*} mice fed HFHC diet for 0 or 16 weeks. Scale bar: 50μm (main panels) and 20μm (Inset). (G) KCs death was quantified. n=4 mice/group. Representative images shown (A, D, F). Unpaired student t-test (B,C,E,G). P value as indicated.

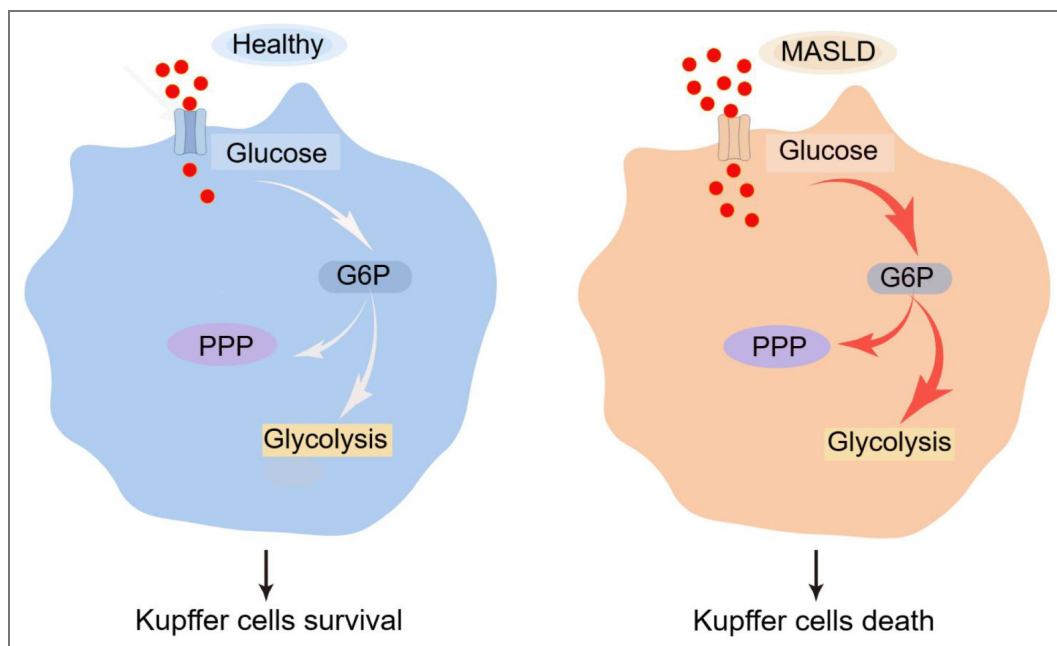


Figure 7. Excessive glycolysis enhancement promotes Kupffer cell death in MASLD.

(Left) Under physiological conditions, Kupffer cells (KCs) maintain basal glucose metabolism supporting cellular homeostasis and survival. **(Right)** During MASLD progression, KCs undergo excessive glycolysis enhancement, which accelerates KCs death.

have described pro-fibrotic functions of Chi3l1, including activation of hepatic stellate cells and modulation of macrophage survival pathways^{28,29,30}. In contrast, using an HFHC model that recapitulates steatohepatitis without fibrosis, we identify a distinct early-stage role for Chi3l1 in maintaining KCs metabolic homeostasis. KC-specific deletion of Chi3l1, but not broader myeloid deletion, accelerated KCs loss and worsened steatosis¹⁷. These results establish Chi3l1 as a KCs-autonomous regulator that restrains glycolytic overload. The apparent duality of Chi3l1 function likely reflects disease stage and cellular source: protective in early MASLD through preservation of resident KCs, but pro-fibrotic in advanced disease when derived from monocyte-derived macrophages and acting on stellate cells.

Our prior work demonstrated that KCs exhibit relative metabolic inflexibility compared with monocyte-derived macrophages, relying heavily on glucose utilization¹⁷. The current study extends this concept by showing that loss of Chi3l1 leads to excessive glycolytic flux that exceeds KCs metabolic tolerance as supported by ¹³C-glucose tracing, enhanced lipotoxic sensitivity *in vitro*, and accelerated cell death *in vivo*. Thus, Chi3l1 functions as a metabolic checkpoint that fine-tunes glucose uptake and prevents lethal glycolytic overload. This selective dependence explains why Chi3l1 deficiency disproportionately affects KCs while sparing other macrophage populations.

Several limitations merit consideration. First, although multiple murine dietary models were used, validation in human MASLD—particularly spatial assessment of KC death and metabolic signatures—is necessary. Second, while we identify glycolytic hyperactivation as an upstream driver, the downstream apoptotic mechanisms remain incompletely defined. Potential contributors include oxidative stress, redox imbalance, lactate accumulation, or mitochondrial insufficiency under high glycolytic flux. Third, the interplay between glucose metabolism and other metabolic pathways, including fatty acid oxidation, warrants further investigation.

Despite these limitations, our findings have translational implications. Therapeutic strategies that preserve KCs metabolic balance—rather than broadly suppressing inflammation—may represent a novel approach for MASLD. Modulation of Chi3l1 signaling or selective targeting of key glycolytic regulators within KCs could help maintain the resident macrophage pool, preserve hepatic immune homeostasis, and potentially delay disease progression. In summary, we identify excessive glycolytic reprogramming as a fundamental mechanism driving selective KC apoptosis in early MASLD, with periportal predominance. Chi3l1 functions as an endogenous metabolic safeguard that restrains this lethal shift. Targeting KCs-specific metabolic vulnerability may offer a new strategy to preserve hepatic immune integrity and modify MASLD progression.

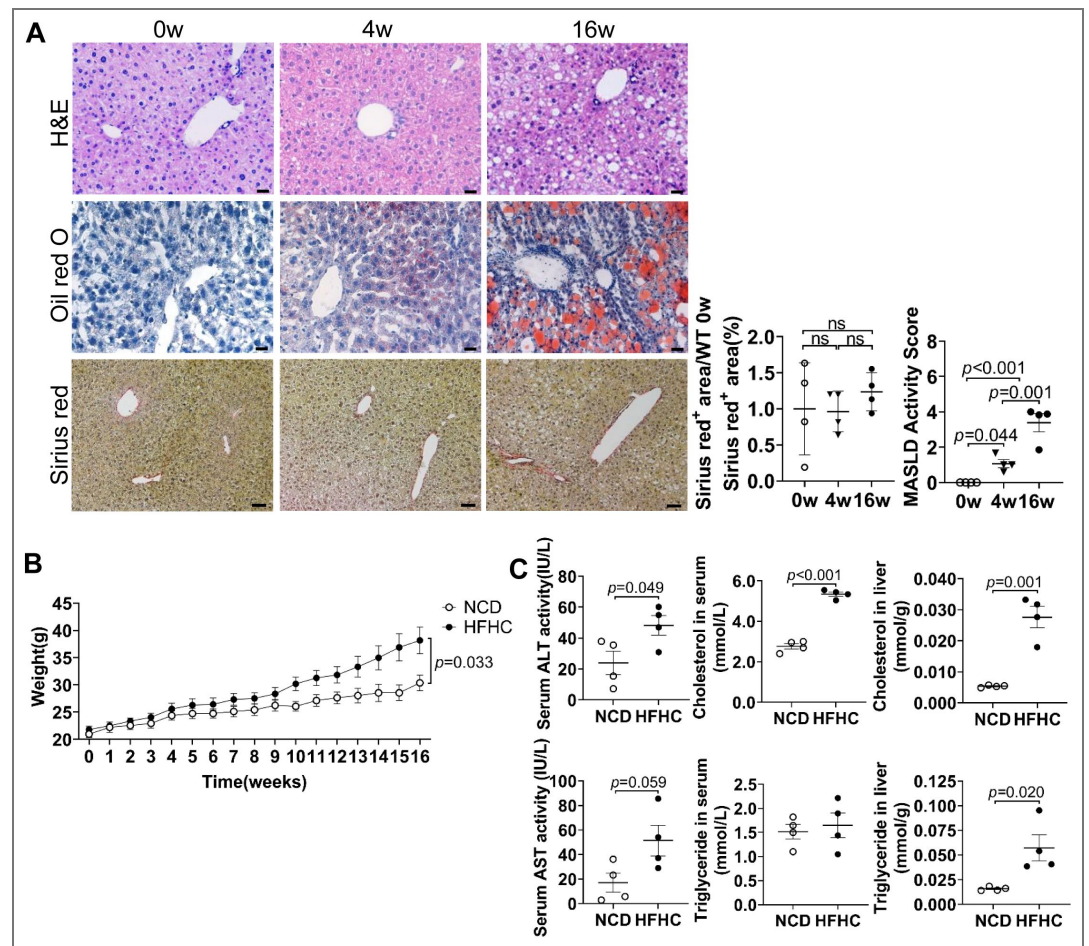


Figure S1. The generation of HFHC-induced MASLD mouse model. Male wildtype C57B/6J mice were fed with HFHC diet for 0, 4 or 16 weeks. **(A)** H&E (top), Oil red O (middle) and Sirius red staining (bottom) were performed to detect MASLD progression in liver sections at 0, 4 or 16 weeks after HFHC diet, respectively. Scale bar: 20µm. Liver fibrosis was quantified. MASLD Activity Score is diagnosed. **(B)** Body weight was recorded during HFHC diet feeding. **(C)** Serum ALT, AST, cholesterol, triglyceride or liver cholesterol, triglyceride is measured at 16 weeks after HFHC diet. n=4 mice/group. Unpaired Student's t-test (B, C); one-way ANOVA (A). P value as indicated. ns: not significant.

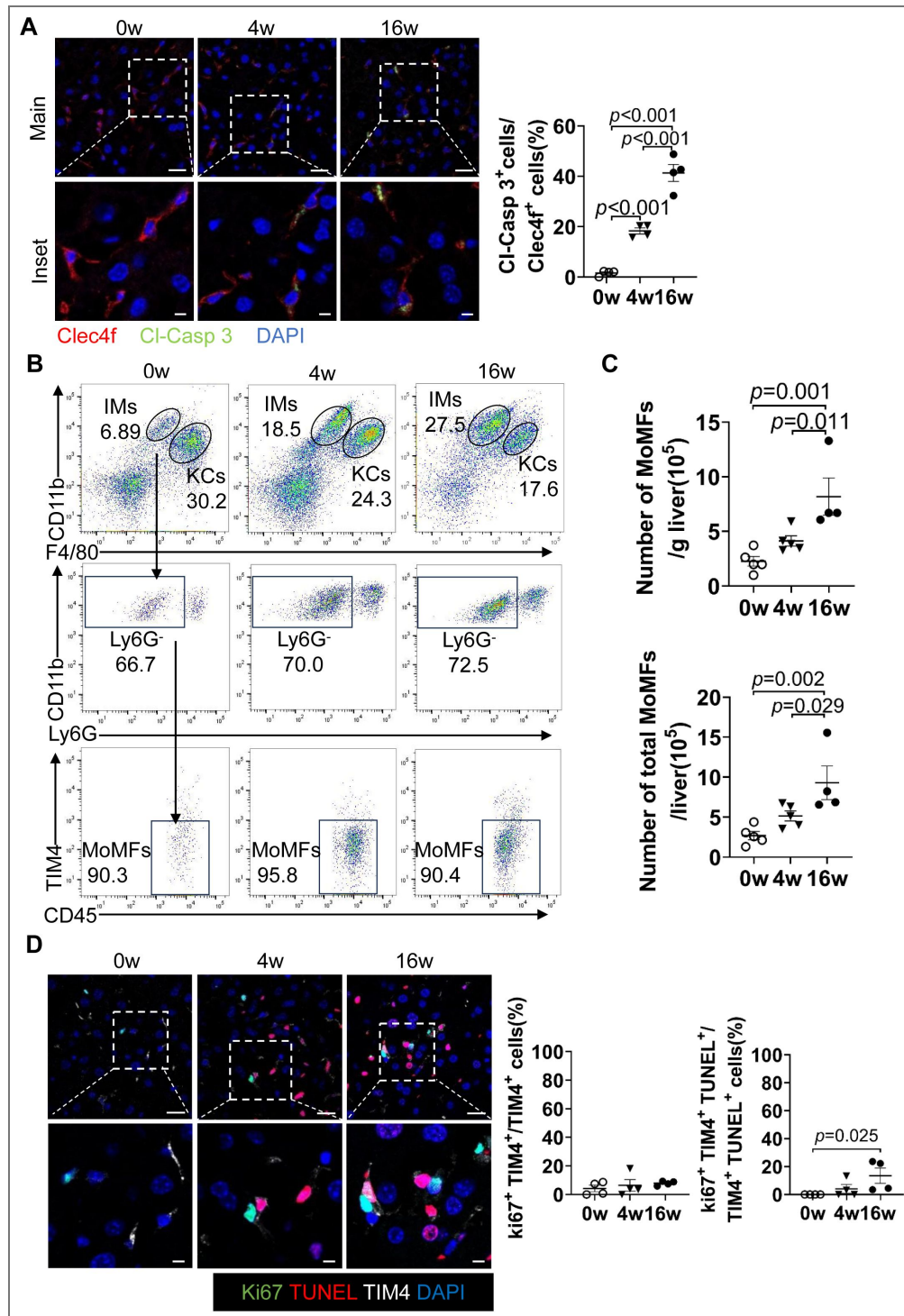


Figure S2. Examination of KCs death and MoMFs recruitment in HFHC mice.

Male wildtype C57B/6j mice were fed with HFHC diet for 0, 4 or 16 weeks. **(A)** KCs death was examined by immunofluorescence staining of Clec4f and cleaved caspase 3 (CI-Casp3) and DAPI (Nuclei) in livers of mice. KCs death was quantified. $n=4$ mice/group. **(B)** Flow cytometry analysis of MoMFs ($CD45^+ Ly6G^- CD11b^+ F4/80^{low} TIM4^{low/-}$) among NPCs of WT mice fed HFHC diet for 0, 4, or 16 weeks. **(C)** MoMFs counts were quantified. $n=4-5$ mice/group. **(D)** Proliferation-associated TUNEL⁺ KCs was examined by co-staining of Ki67, TIM4, TUNEL and DAPI (Nuclei). Ki67⁺KCs among TUNEL⁺ KCs was quantified. Representative images are shown in A, D. One-way ANOVA (A, C, D). P value as indicated. Scale bar: 20 μ m (main panels), 5 μ m(inset).

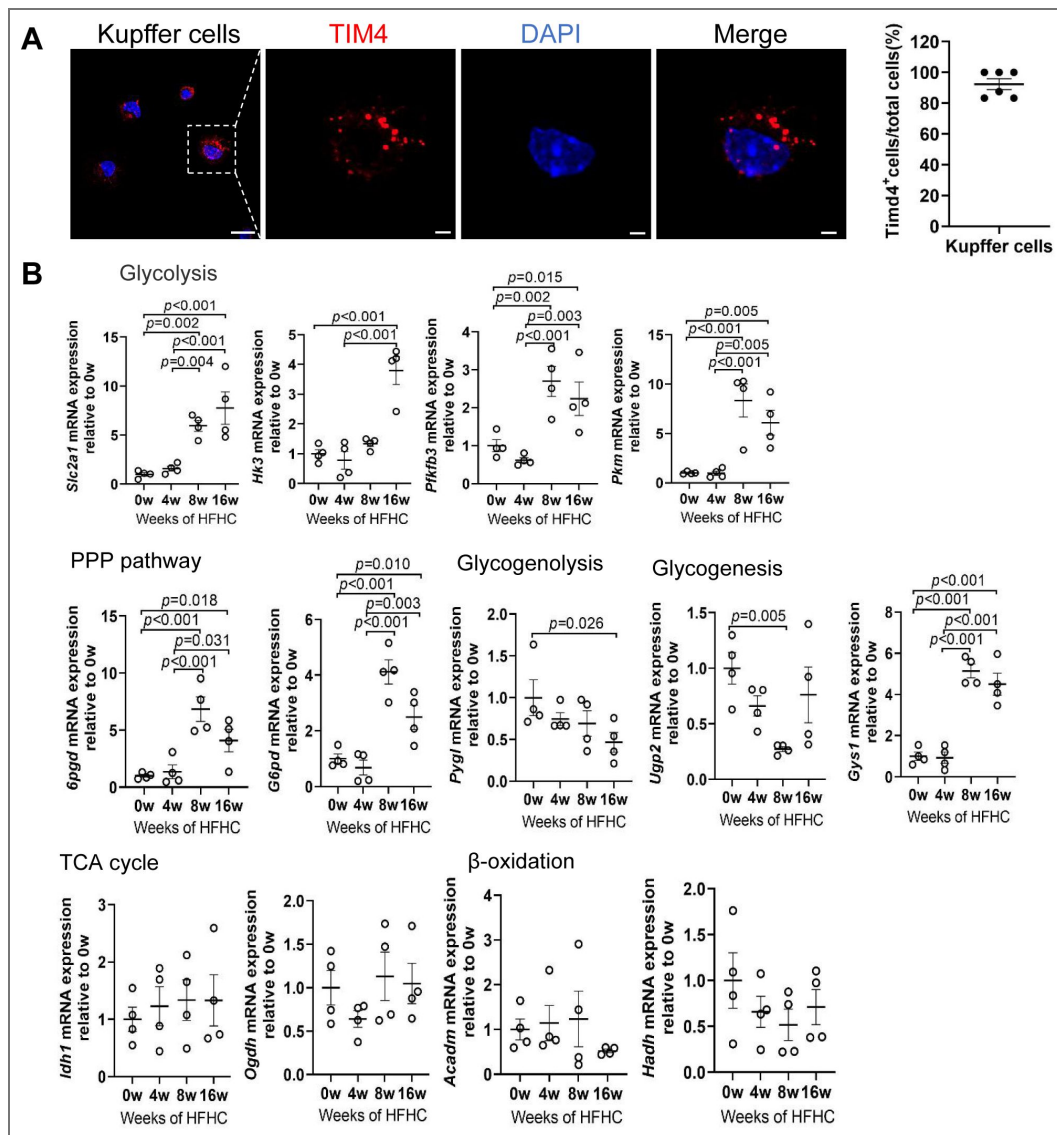


Figure S3. Dynamic changes in mRNA expression levels of rate-limiting enzyme genes involved in glucose metabolism.

(A) Purity of isolated KCs was examined by immunofluorescence staining of TIM4 and DAPI (nuclei). Scale bar:20 μ m (main panels), 5 μ m (inset). Purity was quantified (n=6 independent experiments). (B) qRT-PCR analysis of mRNA expression levels of key rate-limiting enzymes in glycolysis, pentose phosphate pathway (PPP), glycogenolysis, glycogenesis, TCA cycle, and β -oxidation in WT KCs from mice for indicated dietary durations (n = 4 mice/group). One-way ANOVA (B), P values as indicated.

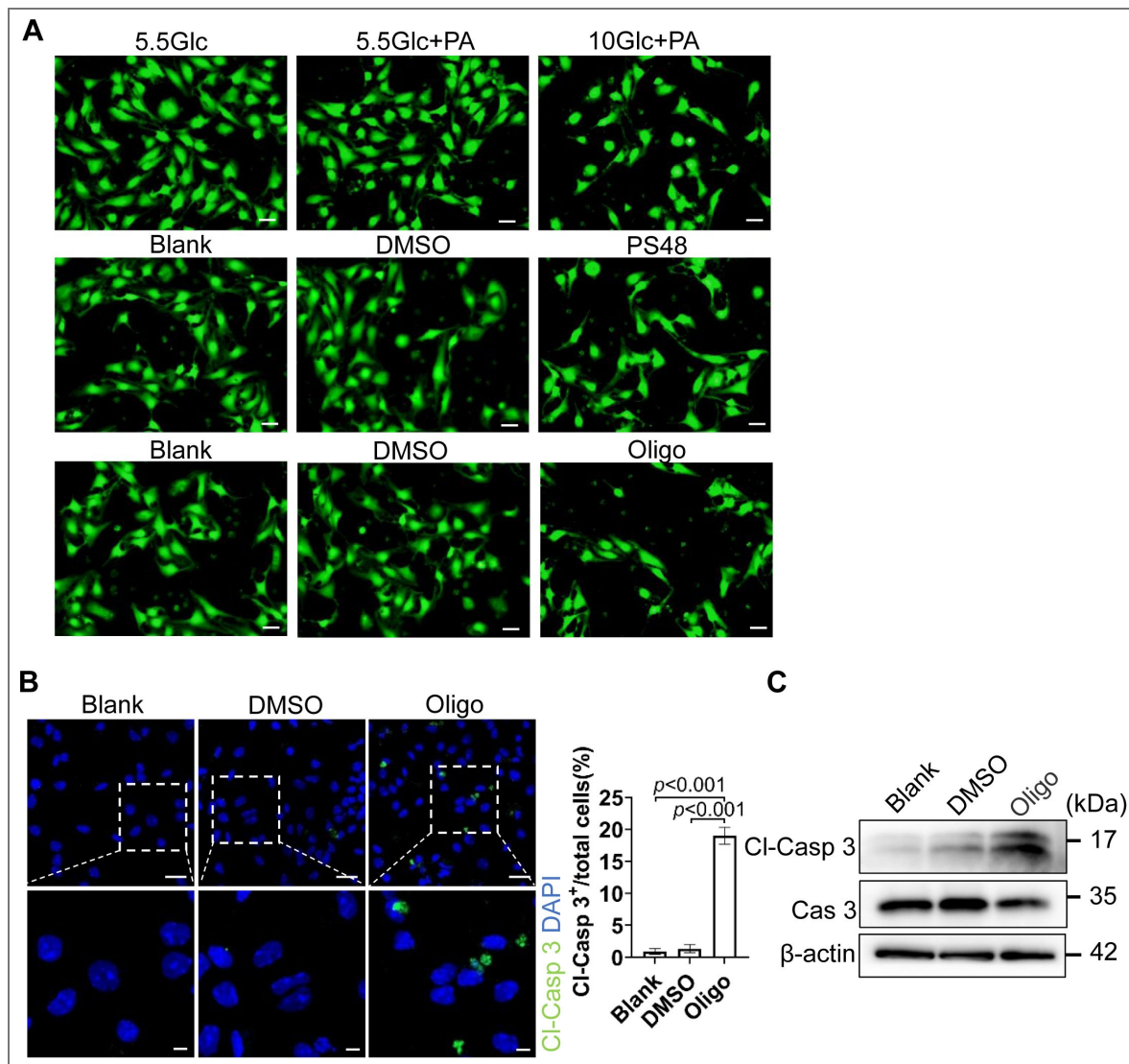


Figure S4. Excessive glucose metabolic activity contributes to Kupffer cell death.

(A) Calcein-AM was used to assess primary KCs cells viability treated as in main Figure 4. Scale bar: 20 μm. Representative images are shown. (B-C) Isolated Kupffer cells were treated for 24 h with: Blank (no treatment), DMSO (vehicle control), 20 μM oligomycin (Oligo, ATP synthase inhibitor). Scale bars: 20 μm (main panels), 5 μm (insets). Cell death were analyzed as above. One-way ANOVA (B). P value as indicated.

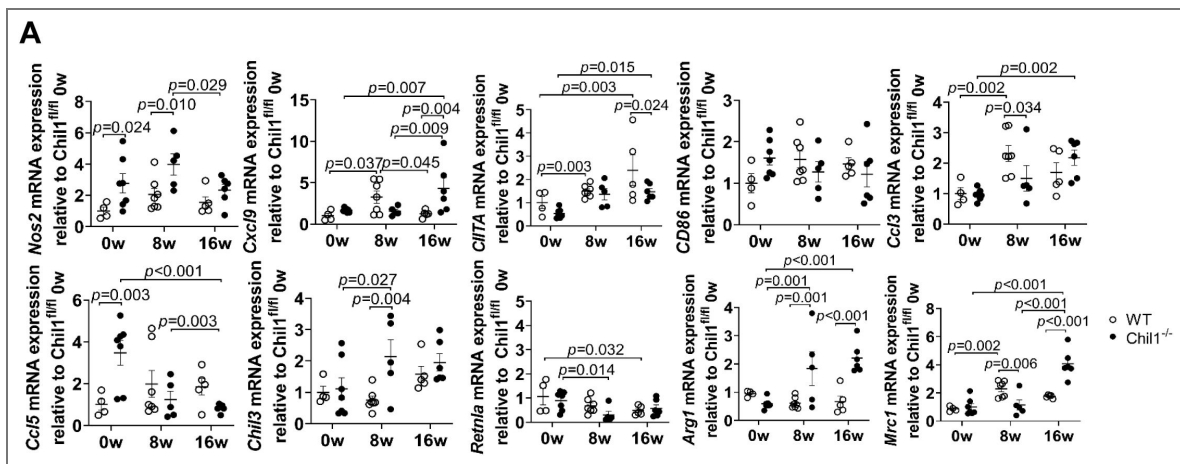


Figure S5. Comparison of KCs polarization between WT and *Chil1^{-/-}* mice during MASLD progression.

(A) qPCR analysis of Key genes involved in macrophage polarization pathway in liver tissues of WT and *Chil1^{-/-}* mice fed with HFHC at indicated week. n=4-7 mice/group. Representative images are shown in A. One-way ANOVA (A). P value as indicated.

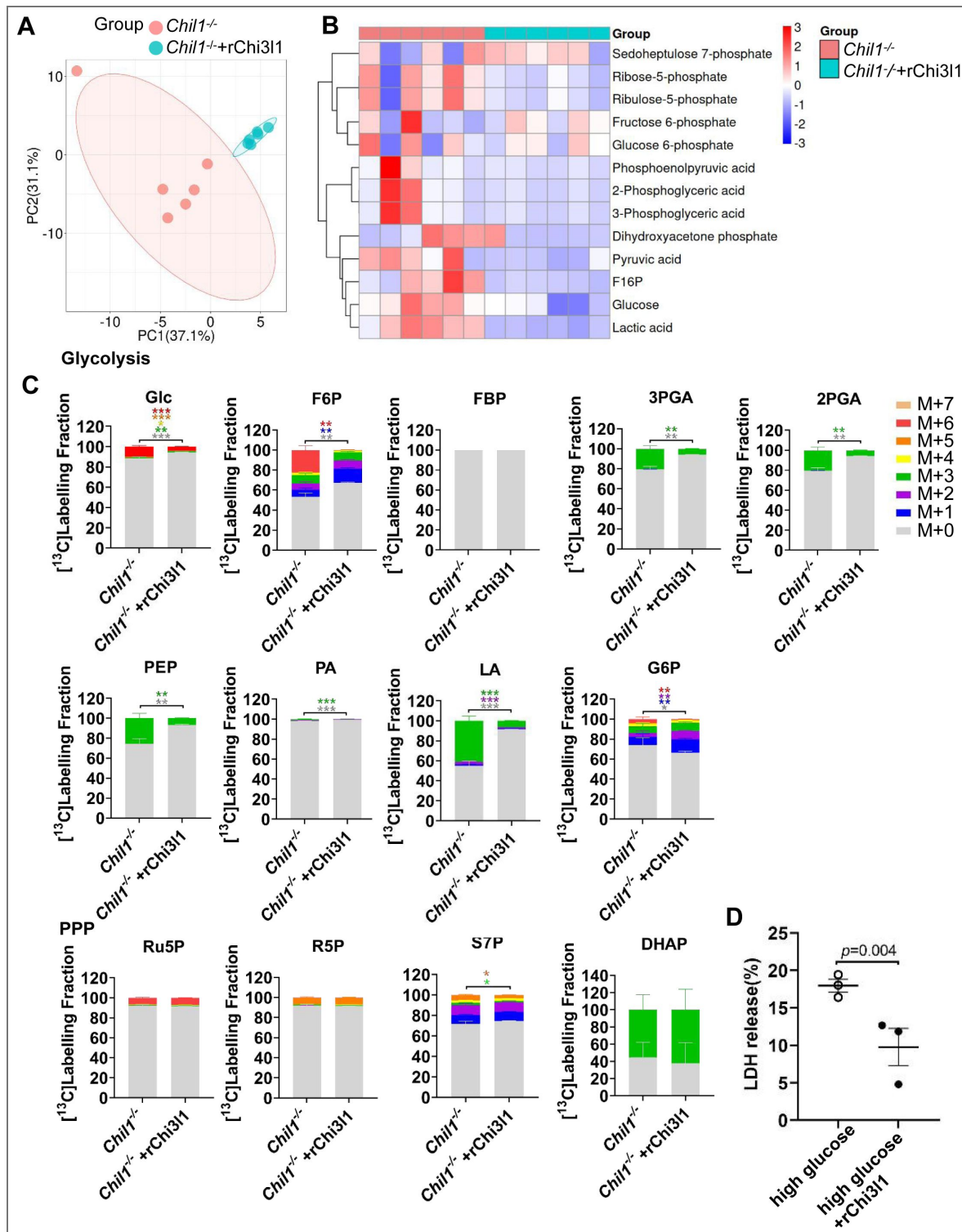


Figure S6. Recombinant Chi3l1 (rChi3l1) inhibits glucose utilization in $\text{Chil1}^{-/-}$ BMDMs.

(A) Comparative principal component analysis (PCA) of metabolites in $\text{Chil1}^{-/-}$ and $\text{Chil1}^{-/-}$ supplemented with rChi3l1 BMDMs cultured with uniformly labeled $[\text{U}-^{13}\text{C}]$ glucose. (B) Heatmap depicting significantly altered Glycolysis and Pentose phosphate (PPP) metabolites in $\text{Chil1}^{-/-}$ and $\text{Chil1}^{-/-}$ supplemented with rChi3l1 BMDMs. (C) Glucose metabolic flux analysis in $\text{Chil1}^{-/-}$ and $\text{Chil1}^{-/-}$ supplemented with rChi3l1 BMDMs showing mass isotopologue distributions of: glycolysis pathway intermediates (Glc, F6P, FBP, 3PGA, 2PGA, PEP, PA, LA, G6P). Pentose phosphate pathway intermediates (Ru5P, R5P, S7P, DHAP). Data represent $n=6$ biological replicates. (D) Lactate dehydrogenase (LDH) activity in culture medium of $\text{Chil1}^{-/-}$ BMDMs treated for 24 h with: 10 mM glucose (high glucose), 10 mM glucose + 100 ng/mL rChi3l1. Unpaired Student's t-test (D). P value as indicated.

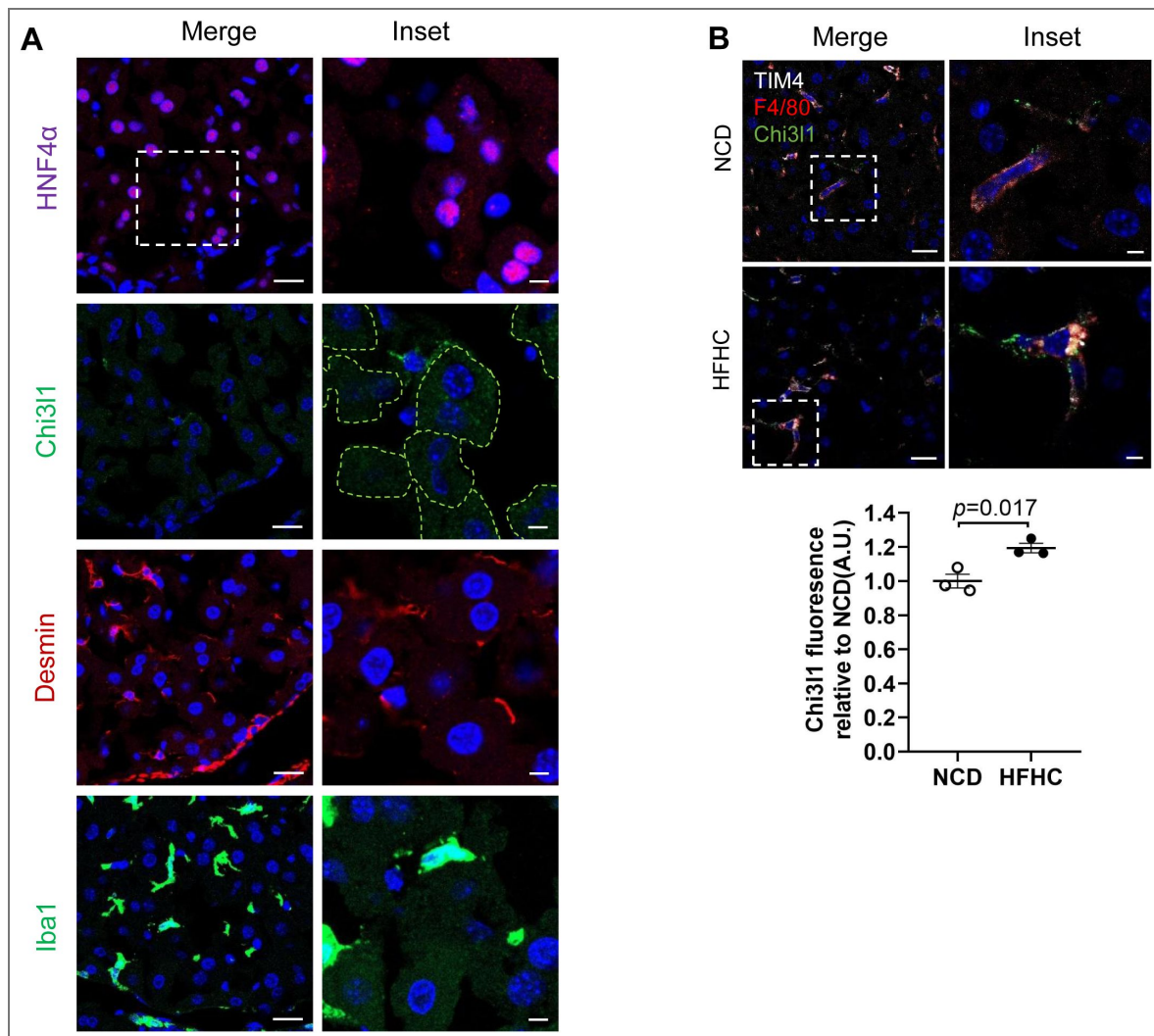


Figure S7. Chi3l1 is majorly expressed in hepatic macrophages and its expression is upregulated during MASLD.

(A) Cellular source of Chi3l1 was assessed by immunohistochemical analysis of consecutive liver sections from mice fed an HFHC diet for 16 weeks. Serial sections were independently stained for: Chi3l1, Lineage markers: HNF4α (hepatocytes), Desmin (hepatic stellate cells; HSCs), or Iba1 (Hepatic macrophages). Cellular localization was determined by aligning morphological profiles across sequential sections. Scale bars: 20 μm (main panels), 5 μm (insets). Hepatocytes are outlined in Chi3l1+ images in green dashed lines. **(B)** Chi3l1 expression in KCs were assessed by co-staining with F4/80 (red) and Timd4 (white) in livers of mice fed either normal chow diet (NCD, 0 weeks) or HFHC diet (16 weeks). Scale bar: 20 μm (main panels), 5 μm (inset). The intensity of Chi3l1 expression in KCs was quantified. Unpaired Student's t-test (B). P value as indicated.

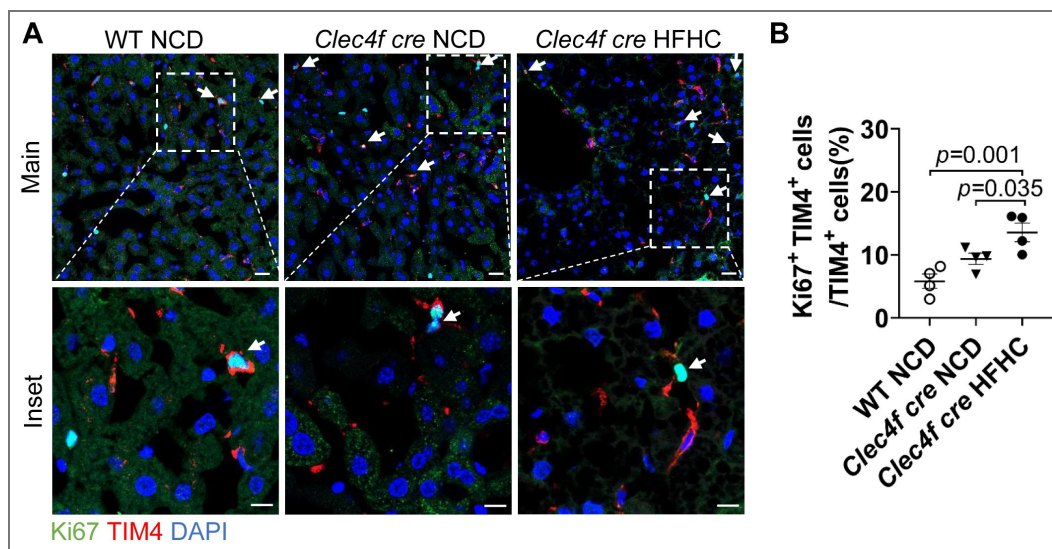


Figure S8. Cre insertion promotes KCs self-proliferation.

(A) Comparison of KCs self-proliferation in WT or *Clec4f cre* mice under NCD or *Clec4f cre* mice under HFHC diet feeding by costaining of Ki67 (proliferation marker) and TIM4 (KCs marker). Nuclei are counterstained with DAPI (blue). Scale bar: 20 μ m (main panels) and 10 μ m (Inset). **(B)** KCs proliferation was quantified. Representative images are shown in A. One-way ANOVA (B). P value as indicated.

Data availability

All data generated or analysed during this study are included in the manuscript and supporting files;source data files have been provided. All reagents developed in this study are available upon reasonable request.

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Additional information

Abbreviations

MASLD, metabolic dysfunction-associated fatty liver disease; MASH, metabolic dysfunction-associated steatohepatitis; KCs, Kupffer cells; Chi3l1, Chitinase 3 like 1; ALT, Alanine aminotransferase; AST, aspartate aminotransferase; TC, cholesterol; TG, triglyceride; NPCs, nonparenchymal cells; HFHC, high fat high cholesterol diet; NCD, normal chow diet; Clec4f, C-type lectin domain family 4; TIM4,T cell immunoglobulin mucin protein 4; MoMFs, monocyte-derived macrophages; HFD, high-fat diet; MCD, methionine/choline deficient diet; WD, western diet; PPP, pentose phosphate pathway; BMDM, bone marrow derived macrophages; DMSO, dimethyl sulfoxide; MAFL, non-alcoholic fatty liver; rChi3l1, recombinant murine Chi3l1; PA, palmitic acid; Iso, Isopropyl alcohol; MoKCs, monocytes-derived Kupffer cells; ALD, alcohol-induced liver disease; AILI, acetaminophen-induced liver injury; EmKCs, embryo-derived Kupffer cells; DT, diphtheria toxin; WT, wild-type; TUNEL, TdT-mediated dUTP Nick-End Labeling; 2-DG, 2-deoxy-D-glucose.

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Author Contributions

JH performed the experiments, analyzed the data, and wrote the manuscript. RL conducted immunofluorescence staining for zonal distribution analysis. CX and XEZ participated in sample collection. KQW assisted with metabolomics analysis. ZS conceived and designed the study, supervised the project, and wrote the manuscript.

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Additional files

[Supplementary materials and methods](#) 

[Supplementary Table 1](#) 

Supplementary Table 2

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Peer reviews

Reviewer #3 (Public review):

This manuscript provides novel insights into altered glucose metabolism and KC status during early MASLD. The authors propose that hyperactivated glycolysis drives a spatially patterned KC depletion that is more pronounced than the loss of hepatocytes or hepatic stellate cells. This concept significantly enhances our understanding of early MASLD progression and KC metabolic phenotype.

Through a combination of TUNEL staining and MS-based metabolomic analyses of KCs from HFHC-fed mice, the authors show increased KC apoptosis alongside dysregulation of glycolysis and the pentose phosphate pathway. Using in vitro culture systems and KC-specific ablation of *Chil1*, a regulator of glycolytic flux, they further show that elevated glycolysis can promote KC apoptosis.

However, it remains unclear whether the observed metabolic dysregulation directly causes KC death or whether secondary factors, such as low-grade inflammation or macrophage activation, also contribute significantly. Nonetheless, the results, particularly those derived from the Chil1-ablated model, point to a new potential target for the early prevention of KC death during MASLD progression.

The manuscript is clearly written and thoughtfully addresses key limitations in the field, especially the focus on glycolytic intermediates rather than fatty acid oxidation. The authors acknowledge the missing mechanistic link between increased glycolysis and KC death. A few things require clarification.

Strengths:

- The study presents the novel observation of profound metabolic dysregulation in KCs during early MASLD and identifies these cells as undergoing apoptosis. The finding that Chil1 ablation aggravates this phenotype opens new avenues for exploring therapeutic strategies to mitigate or reverse MASLD progression.
- The authors provide a comprehensive metabolic profile of KCs following HFHC diet exposure, including quantification of individual metabolites. They further delineate alterations in glycolysis and the pentose phosphate pathway in Chil1-deficient cells, substantiating enhanced glycolytic flux through ¹³C-glucose tracing experiments.
- The data underscore the critical importance of maintaining balanced glucose metabolism in both in vitro and in vivo contexts to prevent KC apoptosis, emphasizing the high metabolic specialization of these cells.
- The observed increase in KC death in Chil1-deficient KCs demonstrates their dependence on tightly regulated glycolysis, particularly under pathological conditions such as early MASLD.

Weaknesses:

- The TUNEL staining in the overview in Figure 2 is not convincing. Typically the signal overlaps with DAPI, which is mostly not the case in the figures shown.
- The mechanistic link between elevated glycolytic flux and KC death remains unclear.
- Figure S5: shows $\Delta\Delta$ CT values, not relative values. What are the housekeeping genes? There should be at least 2, and they should not have metabolically related functions such as Gapdh.
- Figure 1C: shows WT and KO gating side by side
- The following point has not been answered: "While BMDMs from Chil1 knockout mice are used to demonstrate enhanced glycolytic flux, it remains unclear whether Chil1 deficiency affects macrophage differentiation itself." Expression of certain genes that indicate function does not show whether BMDMs isolated from these KO mice are fully differentiated. Here, counting BM input/ BMDM output, flow cytometry on BMDMs, morphology etc. should be tested.

<https://doi.org/10.7554/eLife.109206.2.sa2>

Reviewer #4 (Public review):

Summary:

In this study, He et al. investigate the mechanisms underlying Kupffer cell (KC) loss during metabolic stress. It has long been observed that embryonically derived KCs decline in obesity

and liver disease, a loss that is compensated by monocyte recruitment, although the underlying mechanisms remain unclear. The authors propose that metabolic reprogramming, particularly excessive glycolysis, drives KC death. Using an original murine genetic model to modulate glycolysis, they further demonstrate that enhanced glycolytic activity exacerbates KC damage.

Strengths:

Overall, the study is extremely clearly presented, with a convincing and simple message destined to a vast audience.

Weaknesses:

This manuscript has already undergone one round of revisions in which I was not involved. The authors have tried to address several points raised by the previous reviewers, notably regarding the unexpectedly high level of TUNEL staining observed in KCs. However, I share these concerns expressed by the three reviewers that the reported levels remain difficult to reconcile with the biology. A TUNEL positivity rate of ~60% at week 16 of the HFHC diet would imply massive KC death, which should have led to a near-complete depletion of the KC population, something that is not observed. While I agree that the KC compartment is clearly affected under this dietary challenge, I would strongly encourage the authors to carefully rule out potential technical biases that could account for this implausibly high rate of cell death.

Considering the new in-vivo experiment with 2-DG, it is definitely convincing and clearly adds some value to the full study.

So the full story deserves publication.

<https://doi.org/10.7554/eLife.109206.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors aim to investigate the mechanisms underlying Kupffer cell death in metabolic-associated steatotic liver disease (MASLD). The authors propose that KCs undergo massive cell death in MASLD and that glycolysis drives this process. However, there appears to be a discrepancy between the reported high rates of KC death and the apparent maintenance of KC homeostasis and replacement capacity.

Strengths:

This is an in vivo study.

Weaknesses:

There are discrepancies between the authors' observations and previous reports, as well as inconsistencies among their own findings.

Before presenting the percentage of CLEC4F⁺TUNEL⁺ cells, the authors should have first shown the number of CLEC4F⁺ cells per unit area in Figure 1. At 16 weeks of age, the proportion of TUNEL⁺ KCs is extremely high (~60%), yet the flow cytometry data indicate

that nearly all F4/80⁺ KCs are TIMD4⁺, suggesting an embryonic origin. If such extensive KC death occurred, the proportion of embryonically derived TIMD4⁺ KCs would be expected to decrease substantially. Surprisingly, the proportion of TIMD4⁺ KCs is comparable between chow-fed and 16-week HFHC-fed animals. Thus, the immunostaining and flow cytometry data are inconsistent, making it difficult to explain how massive KC death does not lead to their replacement by monocyte-derived cells.

We thank the reviewer for the insightful comment and the opportunity to clarify this important point. To ensure consistency between our methodologies, we replaced Clec4f staining with TIM4 staining results as requested by the reviewer. We first showed the number of TIM4⁺ cells per unit area in Figure 1B. The results showed a significant and progressive loss of TIM4⁺ cells per unit area in the liver parenchyma, decreasing from approximately 60 cells/FOV at baseline (0w) to nearly 50 at 4w and further to about 30 at 16w post-HFHC diet. This finding is fully consistent with our flow cytometry data. The percentage of the embryonically derived KC population (CD11b^{low} F4/80^{hi} TIM4^{hi}) among CD45⁺ cells dropped from 30.2% (0w) to 24.3% (4w) and 17.6% (16w) (Revised Figure 1C). The absolute number per gram of liver decreased from roughly 12×10^5 (1w) to 9×10^5 (4w) and 5×10^5 (16w) (Revised Figure 1D).

These data suggest that despite the reported high rate of cell death among CLEC4F⁺TIMD4⁺ KCs, the population appears to self-maintain, with no evidence of monocyte-derived KC generation in this model, which contradicts several recent studies in the field.

We appreciate the reviewer's insightful comment. We agree that our data show no substantial generation of monocyte-derived Kupffer cells (MoKCs) within the 16-week HFHC model. However, we do not believe the remaining embryonic KCs (EmKCs) are maintained through self-renewal, as the proportion of Ki67⁺TIM4⁺ cells remains low at all time points (Revised Figure S2D). Instead, our observations align with a phased replacement model: recruited monocytes first differentiate into monocyte-derived macrophages (MoMFs), which we see accumulate (Revised Figure S2B, S2C), and only later adopt a KC phenotype. Consistent with this, our 16-week model shows significant EmKC loss and MoMF expansion, but not yet the emergence of TIM4-MoKCs. This timing is supported by prior studies, where TIM4-KCs were observed at 24 weeks, but not at 16 weeks, on similar diets (Ref. 1,2). Therefore, we interpret our findings as capturing an earlier phase of MASLD progression, characterized by EmKC death and MoMF accumulation, prior to their full differentiation into MoKCs.

Moreover, there is no evidence that TIM4⁺CLEC4F⁺ KCs increase their proliferation rate to compensate for such extensive cell death. If approximately 60% of KCs are dying and no monocyte-derived KCs are recruited, one would expect a much greater decrease in total KC numbers than what is reported.

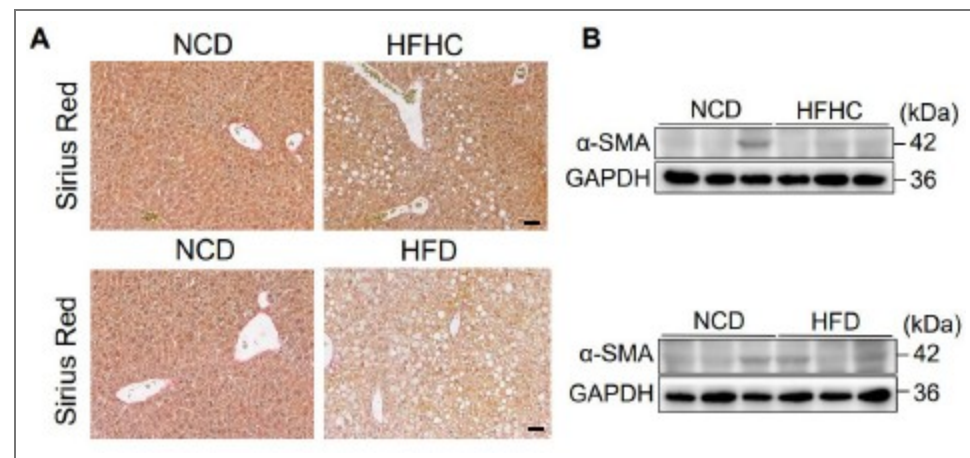
Thank you for raising this point, which allows for an important clarification. The interpretation that approximately 60% of KCs are dying is correct, but this refers to the proportion of the remaining KC population at 16 weeks that is TUNEL⁺, not to 60% of the original KC pool. Since our data show that over half of the EmKCs are lost by 16 weeks (Revised Figure 1B), the 60% of dying cells at this late time point corresponds roughly to only 25-30% of the total original KC population at baseline. This distinction reconciles the high rate of apoptosis observed late in disease with the overall progressive depletion of the EmKC pool.

It is also unexpected that the maximal rate of KC death occurs at early time points (8 weeks), when the mice have not yet gained substantial weight (Figure 1B). Previous studies have shown that longer feeding periods are typically required to observe the loss of embryo-derived KCs.

We appreciate the reviewer's insightful observation. We think KC death is a continuous event during MASLD. To induce MASH, previous studies typically assess the loss of EmKCs after longer feeding periods, which might leave us an impression that longer feeding periods are required to observe substantive loss of embryonically derived KCs. In our HFHC model, the proportion of dying KCs was already elevated by 8 weeks, and this high rate was sustained through the 16-week endpoint. In a separate MCD dietary model characterized by rapid MASLD progression, a high rate of KC death was detectable as early as 6 weeks (Revised Figure 1F). Collectively, these data suggest that the onset of significant KC death is dependent on the pace of MASLD pathogenesis, more likely an early-initiated event that is through MASLD progression.

Furthermore, it is surprising that the HFD induces as much KC death as the HFHC and MCD diets. Earlier studies suggested that HFD alone is far less effective than MASH-inducing diets at promoting the replacement of embryonic KCs by monocyte-derived macrophages.

We appreciate the reviewer's insightful comment. In our study, we observed significant KCs death under both HFD and HFHC feeding for 20, 16 weeks, respectively. Moreover, both HFHC and HFD induced similar stages of MASLD (characterized by significant lipid accumulation without fibrosis development) by these time points (Authir response image 1). Therefore, these data support that the onset of substantial KCs death may be an early MASLD event, before the progression to MASH. Additionally, this finding aligns with existing literature showing that 16 weeks of HFD feeding alone is sufficient to cause a marked reduction in the TIM4⁺KCs population (Ref. 1).



Author response image 1. Detection of liver fibrosis in MASLD mouse models. Male wild-type C57BL/6J mice were fed a high-fat, high-cholesterol (HFHC) diet for 16 weeks or a high-fat diet (HFD) for 20 weeks to induce MASLD. Mice fed a normal chow diet (NCD) served as controls. (A) Sirius Red staining of liver sections was performed to assess collagen deposition and fibrosis during MASLD progression. Scale bar, 20 μ m. (B) Western blot analysis of liver tissue lysates showing α -smooth muscle actin (α -SMA) expression as a marker of hepatic stellate cell activation and liver fibrosis.

In Figure 2D, TIMD4 staining appears extremely faint, making the results difficult to interpret. In contrast, the TUNEL signal is strikingly intense and encompasses a large proportion of liver cells (approximately 60% of KCs, 15% of hepatocytes, 20% of hepatic stellate cells, 30% of non-KC macrophages, and a proportion of endothelial cells is also likely affected). This pattern closely resembles that typically observed in mouse models of acute liver failure. Given this apparent extent of cell death, it is unexpected that ALT and AST levels remain low in MASH mice, which is highly unusual.

Thank you for this important feedback. To address concerns about the clarity of our imaging, we have provided high-resolution split-channel raw images for Figure 2D (Revised Figure 2D), which distinctly show the localization of TIM4, TUNEL, and GS. These confirm the progressive reduction of TIM4⁺KCs and the increase in TUNEL⁺ TIM4⁺ cells over time. We agree that the high proportion of TUNEL⁺ cells seems at odds with the modest ALT/AST elevation. This discrepancy might be explained by the distinct nature of cell death in MASLD. Unlike the acute necrosis with membrane rupture seen in acute liver failure—which causes massive, rapid enzyme release—obesity-related liver injury is a chronic process dominated by apoptosis (Ref. 4,5). Apoptosis preserves membrane integrity until late stages (Ref. 6), with dying cells packaged into apoptotic bodies for efficient phagocytic clearance by neighboring macrophages (Ref. 7,8). This controlled disposal system minimizes the leakage of intracellular enzymes. Therefore, the coexistence of widespread apoptosis (high TUNEL signal) with limited enzyme release (low ALT/AST) is a recognized feature of chronic MASLD pathogenesis.

No statistical analysis is provided for Figure 5D, and it is unclear which metabolites show statistically significant changes in Figure 5C.

We thank the reviewer for raising this statistical problem. We have now included statistical analysis in Revised Figure 5D.

In addition, there is no evaluation of liver pathology in Clec4f-Cre × Chil1flox/flox mice. It remains possible that the observed effects on KC death result from aggravated liver injury in these animals. There is also no evidence that Chil1 deficiency affects glucose metabolism in KCs in vivo.

We thank the reviewer for these important points. We previously characterized the liver pathology of Clec4f^{ΔChil1} mice in detail (preprint: eLife 2025, DOI: 10.7554/eLife.107023.1, Fig. 2). On a normal chow diet, these mice showed no differences in body weight, hepatic lipid deposition, metabolic parameters, or glucose tolerance compared to controls. However, on an HFHC diet, Clec4f^{ΔChil1} mice developed significantly worse metabolic and histological phenotypes. Crucially, our *in vitro* data demonstrate that recombinant Chi3l1 directly reduces KC death (preprint, Fig. 6E-F), indicating that the aggravated MASLD in knockout mice is a consequence of increased KC loss, not its cause.

Regarding glucose metabolism, we have previously shown that Chi3l1 deficiency leads to increased glucose uptake by KCs *in vivo* using the fluorescent glucose analog 2-NBDG. This effect was reversed by supplementing knockout mice with recombinant Chi3l1 (preprint Fig. 6G-H). This provides direct evidence that Chi3l1 modulates glucose uptake in KCs *in vivo*.

Finally, the authors should include a more direct experimental approach to modulate glycolysis in KCs and assess its causal role in KC death in MASH.

We thank the reviewer for this constructive suggestion. To more directly evaluate the role of glycolysis in KCs death *in vivo*, we performed pharmacological inhibition of glycolysis using 2-deoxy-D-glucose (2-DG) in the HFHC-induced MASLD model (Revised Figure 4E–G). Wild-type mice were fed an HFHC diet for four weeks, and 2-DG (50 mg/kg) or vehicle was administered intraperitoneally every other day beginning at week 3. This short intervention period and modest dosing were chosen to limit potential systemic metabolic effects while modulating glycolytic activity during active disease development. KCs apoptosis was assessed by TIM4/TUNEL co-staining. 2-DG treatment significantly reduced the proportion of TUNEL⁺KCs compared with vehicle controls, indicating protection against KCs death. These data together with our complementary *in vitro* gain-of-function experiments, support a contributory role for excessive glycolytic activity in promoting KC apoptosis in MASLD. We have incorporated these findings into the revised manuscript to strengthen the causal link

between glycolytic reprogramming and KCs loss *in vivo* (Revised manuscript, page 7, line 267-282).

Reviewer #2 (Public review):

Summary:

In this manuscript, He et al. set out to investigate the mechanisms behind Kupffer Cell death in MASLD. As has been previously shown, they demonstrate a loss of resident KCs in MASLD in different mouse models. They then go on to show that this correlates with alterations in genes/metabolites associated with glucose metabolism in KCs. To investigate the role of glucose metabolism further, they subject isolated KCs in vitro to different metabolic treatments and assess cleaved caspase 3 staining, demonstrating that KCs show increased Cl. Casp 3 staining upon stimulation of glycolysis. Finally, they use a genetic mouse model (Chil1KO) where they have previously reported that loss of this gene leads to increased glycolysis and validate this finding in BMDMs (KO). They then remove this gene specifically from KCs (Clec4fCre) and show that this leads to increased macrophage death compared with controls.

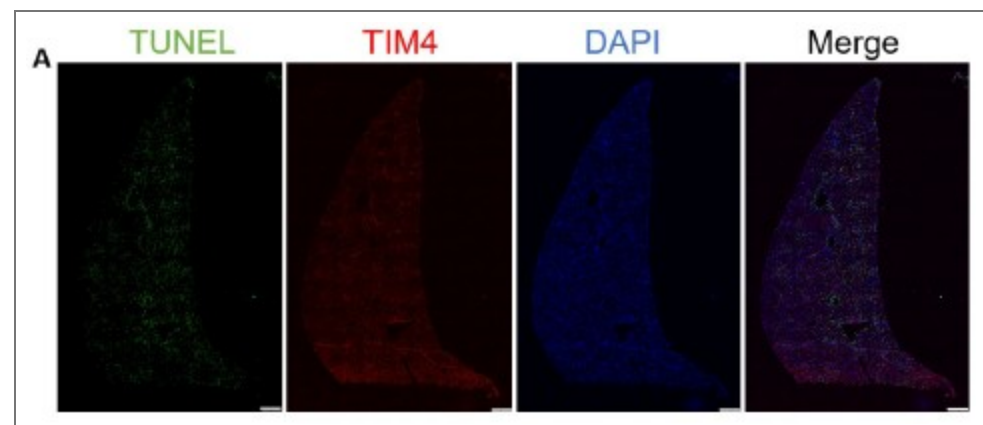
Strengths:

As we do not yet understand why KCs die in MASLD, this manuscript provides some explanation for this finding. The metabolomics is novel and provides insight into KC biology. It could also lead to further investigation; here, it will be important that the full dataset is made available.

Weaknesses:

Different diets are known to induce different amounts of KC loss, yet here, all models examined appear to result in 60% KC death. One small field of view of liver tissue is shown as representative to make these claims, but this is not sufficient, as anything can be claimed based on one field of view. Rather, a full tissue slice should be included to allow readers to really assess the level of death.

Thank you for raising this point regarding data presentation. We analyzed full tissue slices and found that including a view of the entire slice at a standard magnification makes individual KC difficult to resolve (Author response image 2). To clearly represent the extent and distribution of KCs death across the liver tissue slice, we now include lower-magnification images that provide a wider field of view, allowing readers to assess the pattern across a larger tissue area (Revised Figures 1, 2, 6F).



Author response image 2. Assessment of KCs death on full liver tissue slice. (A) Immunofluorescence staining was performed to detect Kupffer cell (KC) death in liver sections from mice fed an MCD diet for 6 weeks. Cell death

was assessed by TUNEL staining (green), and KCs were identified by TIM4 staining (red). Nuclei were counterstained with DAPI (blue). Representative whole-tissue view is shown. Scale bars, 1mm.

Additionally, there is no consistency between the markers used to define KCs and moMFs, with CLEC4F being used in microscopy, TIM4 in flow, while the authors themselves acknowledge that moKCs are CLEC4F+TIM4-. As moKCs are induced in MASLD, this limits interpretation. Additionally, Iba1 is referred to as a moMF marker but is also expressed by KCs, which again prevents an accurate interpretation of the data. Indeed, the authors show 60% of KCs are dying but only 30% of IBA1+ moMFs, as KCs are also IBA1+, this would mean that KCs die much more than moMFs, which would then limit the relevance of the BMDM studies performed if the phenotype is KC specific. Therefore, this needs to be clarified.

We thank the reviewer for the constructive comments. For consistency, we have standardized our KC marker to TIM4 for all immunostaining data, aligning it with our flow cytometry analysis (Revised Figures 1, 2D, 6F). We have also clarified that IBA1 is expressed by hepatic macrophages (both KCs and MoMFs)(Revised Figure 2C, Revised manuscript, page 5, lines 182-183). Moreover, we also included the clarification that 60% of TIM4⁺ KCs are TUNEL⁺ versus 30% of total IBA1⁺ cells further supports that KCs undergo death more readily than MoMFs (Revised manuscript, page 5, lines 186-189). We also acknowledged the limitation of BMDM studies in the Revised manuscript, page 8, line 332-340.

The claim that periportal KCs die preferentially is not supported, given that the majority of KCs are peri-portal. Rather, these results would need to be normalised to KC numbers in PP vs PC regions to make meaningful conclusions.

We thank the reviewer for this important point. We included the normalized data. At 8 weeks, the normalized death rate was significantly higher in periportal versus pericentral regions ($p = 0.041$), supporting increased periportal KC susceptibility during early MASLD. By 16 weeks, proportional death rates became comparable between zones (Revised Figure 2D, Revised manuscript, page 6, lines 194-201).

Additionally, KCs are known to be notoriously difficult to keep alive in vitro, and for these studies, the authors only examine cl. Casp 3 staining. To fully understand that data, a full analysis of the viability of the cells and whether they retain the KC phenotype in all conditions is required.

We appreciate the reviewer's suggestions. To confirm the identity and health of isolated KCs in our *in vitro* studies, we showed that ~95% of primary isolated KCs are TIM4⁺ (Revised Figure S3A). Furthermore, Calcein-AM staining confirmed that the remaining KCs under our experimental conditions are viable and healthy (Revised Figure S4A).

Finally, in the Cre-driven KO model, there does not seem to be any death of KCs in the controls (rather numbers trend towards an increase with time on diet, Figure 6E), contrary to what had been claimed in the rest of the paper, again making it difficult to interpret the overall results.

We thank the reviewer for this comment. During our analysis, we indeed observed no reduction in KCs in the Clec4f cre control mice. This prompted us to consider that Cre insertion itself might influence KCs maintenance. To investigate this, we performed TIM4/Ki67 co-staining, which revealed significantly higher numbers of proliferating KCs in Clec4f cre mice compared with C57BL/6J mice under NCD. Following HFHC feeding, KCs proliferation in Clec4f cre mice increased even further. These results indicate that Cre insertion enhanced KCs self-renewal in Clec4f cre mice, which contributes to maintenance of the KCs pool during MASLD (Revised Figures S8A and S8B). (Revised manuscript, page 9, line 363-370).

Additionally, there is no validation that the increased death observed *in vivo* in KCs is due to further promotion of glycolysis.

We thank the reviewer for this constructive suggestion. To more directly evaluate the role of glycolysis in KCs death *in vivo*, we performed pharmacological inhibition of glycolysis using 2-deoxy-D-glucose (2-DG) (Revised Figure 4E–G). Wild-type mice were fed an HFHC diet for five weeks, and 2-DG (50 mg/kg) or vehicle was administered intraperitoneally every other day beginning at week 3. This short intervention period and modest dosing were chosen to limit potential systemic metabolic effects while modulating glycolytic activity in KCs. KCs apoptosis was assessed by TIM4/TUNEL co-staining. 2-DG treatment significantly reduced the proportion of TUNEL⁺ KCs compared with vehicle controls, indicating protection against KCs death. These data, together with our complementary *in vitro* gain-of-function experiments support a contributory role for excessive glycolytic activity in promoting KCs death in MASLD. We have incorporated these findings into the revised manuscript to strengthen the causal link between glycolytic reprogramming and KCs loss *in vivo* (Revised manuscript, page 7, line 267-282).

Reviewer #3 (Public review):

This manuscript provides novel insights into altered glucose metabolism and KC status during early MASLD. The authors propose that hyperactivated glycolysis drives a spatially patterned KC depletion that is more pronounced than the loss of hepatocytes or hepatic stellate cells. This concept significantly enhances our understanding of early MASLD progression and KC metabolic phenotype.

*Through a combination of TUNEL staining and MS-based metabolomic analyses of KCs from HFHC-fed mice, the authors show increased KC apoptosis alongside dysregulation of glycolysis and the pentose phosphate pathway. Using *in vitro* culture systems and KC-specific ablation of *Chil1*, a regulator of glycolytic flux, they further show that elevated glycolysis can promote KC apoptosis.*

*However, it remains unclear whether the observed metabolic dysregulation directly causes KC death or whether secondary factors, such as low-grade inflammation or macrophage activation, also contribute significantly. Nonetheless, the results, particularly those derived from the *Chil1*-ablated model, point to a new potential target for the early prevention of KC death during MASLD progression.*

The manuscript is clearly written and thoughtfully addresses key limitations in the field, especially the focus on glycolytic intermediates rather than fatty acid oxidation. The authors acknowledge the missing mechanistic link between increased glycolysis and KC death. Still, several interpretations require moderation to avoid overstatement, and certain experimental details, particularly those concerning flow cytometry and population gating, need further clarification.

Strengths:

*(1) The study presents the novel observation of profound metabolic dysregulation in KCs during early MASLD and identifies these cells as undergoing apoptosis. The finding that *Chil1* ablation aggravates this phenotype opens new avenues for exploring therapeutic strategies to mitigate or reverse MASLD progression.*

*(2) The authors provide a comprehensive metabolic profile of KCs following HFHC diet exposure, including quantification of individual metabolites. They further delineate alterations in glycolysis and the pentose phosphate pathway in *Chil1*-deficient cells, substantiating enhanced glycolytic flux through ¹³C-glucose tracing experiments.*

(3) The data underscore the critical importance of maintaining balanced glucose metabolism in both in vitro and in vivo contexts to prevent KC apoptosis, emphasizing the high metabolic specialization of these cells.

(4) The observed increase in KC death in *Chil1*-deficient KCs demonstrates their dependence on tightly regulated glycolysis, particularly under pathological conditions such as early MASLD.

Weaknesses:

(1) The novelty is questionable. The presented work has considerable overlap with a study by the same lab, which is currently under review (citation 17), and it should be considered whether the data should not be presented in one paper.

We appreciate the reviewer for the opportunity to clarify the relationship between the two studies. In our previous work (citation 17), we focused on the transcriptional metabolic differences between Kupffer cells (KCs) and monocyte-derived macrophages (MoMFs) and identified *Chi3l1* as a selective protective factor that limits glucose uptake and shields KCs from metabolic stress-induced cell death, with minimal effects on MoMFs. That study directly motivated the current work. The observation that KCs are uniquely protected from metabolic stress led us to hypothesize that excessive glycolytic activation itself may be a primary driver of KCs death, which forms the central question of the present study. Accordingly, the current manuscript shifts the focus from *Chi3l1*-mediated protection to the mechanistic role of hyperglycolysis in driving KCs mortality, using distinct experimental approaches and addressing a different biological question. Because the two studies address conceptually distinct aims—one defining a protective regulator of KCs survival and the other dissecting glycolysis-driven KCs death mechanisms—we believe they are best presented as separate manuscripts. Combining them into a single study would dilute the mechanistic depth and clarity of each story.

(2) The authors report that 60% of KCs are TUNEL-positive after 16 weeks of HFHC diet and confirm this by cleaved caspase-3 staining. Given that such marker positivity typically indicates imminent cell death within hours, it is unexpected that more extensive KC depletion or monocyte infiltration is not observed. Since *Timd4* expression on monocyte-derived macrophages takes roughly one month to establish, the authors should consider whether these TUNEL-positive KCs persist in a pre-apoptotic state longer than anticipated. Alternatively, fate-mapping experiments could clarify the dynamics of KC death and replacement.

We thank the reviewer for this astute observation. As shown in revised Figure 2D, the proportion of *TIM4*⁺*TUNEL*⁺KCs peaks at 8 weeks after HFHC feeding and remains elevated at 16 weeks. However, examination of the corresponding single-channel *TIM4* staining during this period reveals that the overall density of *TIM4*⁺ KCs does not undergo abrupt or synchronous depletion. This temporal dissociation between sustained TUNEL positivity and relatively gradual KCs loss suggests that TUNEL-positive KCs do not undergo immediate clearance. Based on these observations, we agree with the reviewer that a substantial fraction of TUNEL-positive KCs likely persists in a prolonged pre-apoptotic or stressed state rather than undergoing rapid cell death. This interpretation is consistent with the absence of extensive KCs depletion or compensatory monocyte infiltration at these time points. Importantly, previous studies (Ref. 1,2) indicate that KCs are eventually lost as MASLD progresses, supporting the notion that KC death is a gradual process that unfolds over an extended time frame rather than acutely.

(3) The mechanistic link between elevated glycolytic flux and KC death remains unclear.

We thank the reviewer for this constructive suggestion. To more directly evaluate the role of glycolysis in KCs death *in vivo*, we performed pharmacological inhibition of glycolysis using 2-deoxy-D-glucose (2-DG) (Revised Figure 4E–G). Wild-type mice were fed an HFHC diet for five weeks, and 2-DG (50 mg/kg) or vehicle was administered intraperitoneally every other day beginning at week 3. This short intervention period and modest dosing were chosen to limit potential systemic metabolic effects while modulating glycolytic activity of KCs. KCs apoptosis was assessed by TIM4/TUNEL co-staining. 2-DG treatment significantly reduced the proportion of TUNEL⁺KCs compared with vehicle controls, indicating protection against KCs death. These data, together with our complementary *in vitro* gain-of-function experiments, support a contributory role for excessive glycolytic activity in promoting KC apoptosis in MASLD. We have incorporated these findings into the revised manuscript to strengthen the causal link between glycolytic reprogramming and KCs loss *in vivo* (Revised manuscript, page 7, line 267-282).

(4) The study does not address the polarization or ontogeny of KCs during early MASLD. Given that pro-inflammatory macrophages preferentially utilize glycolysis, such data could provide valuable insight into the reason for increased KC death beyond the presented hyperreliance on glycolysis.

We thank the reviewer for this insightful comment. Regarding KCS ontogeny, flow cytometry analysis (Revised Figure 1C) shows that KCs remain uniformly TIM4^{hi} during early MASLD, indicating that monocyte-derived KCs (TIM4^{low}) have not yet emerged at these stages. To address KCs polarization, we assessed the expression of M1-type (pro-inflammatory) markers (Nos2, Cxcl9, CIITA, Cd86, Ccl3, and Ccl5) and M2-type (anti-inflammatory) markers (Chil3, Retnla, Arg1, and Mrc1) in KCs isolated from WT mice fed a HFHC diet for 0, 8, and 16 weeks. As shown in revised Figure S5A, M1 markers progressively increase over time, whereas M2 markers remain unchanged or slightly decrease. This polarization shift is consistent with the increased glycolytic activity observed in KCs during early MASLD. Together, these data indicate that embryonically derived KCs undergo a pro-inflammatory polarization accompanied by enhanced glycolytic metabolism during early MASLD, providing mechanistic context for their increased susceptibility to metabolic stress-induced cell death beyond hyperreliance on glycolysis alone (Revised manuscript, page 7-8, line 307-321).

(5) The gating strategy for monocyte-derived macrophages (moMFs) appears suboptimal and may include monocytes. A more rigorous characterization of myeloid populations by including additional markers would strengthen the study's conclusions.

We thank the reviewer for raising this important point. To improve the rigor of our analysis, we adopted gating strategies established in previous studies (PMID: 41131393; PMID: 32562600). Specifically, Kupffer cells were defined as CD45⁺CD11b⁺F4/80^{hi} TIM4^{hi} cells, while monocyte-derived macrophages (MoMFs) were defined as CD45⁺Ly6G[−]CD11b⁺F4/80^{low} TIM4^{low/−} cells, thereby excluding contaminating neutrophils and minimizing inclusion of circulating monocytes. Using this refined gating strategy, we observed a progressive reduction of KCs accompanied by a corresponding increase in MoMFs in WT mice during HFHC feeding (Revised Figures 1C and S2B–C), (Revised manuscript, page 4, line 154-163).

(6) While BMDMs from Chil1 knockout mice are used to demonstrate enhanced glycolytic flux, it remains unclear whether Chil1 deficiency affects macrophage differentiation itself.

We thank the reviewer for this important question. To determine whether Chi3l1 deficiency affects macrophage differentiation, we analyzed the expression of M1-type (pro-inflammatory) markers (Nos2, Cxcl9, CIITA, Cd86, Ccl3, and Ccl5) and M2-type (anti-inflammatory) markers (Chil3, Retnla, Arg1, and Mrc1) in Kupffer cells isolated from WT and *Chil1*^{−/−} mice fed a HFHC diet for 0, 8, and 16 weeks. At baseline (0 weeks), Chi3l1 deficiency was associated with elevated expression of multiple M1 markers, whereas M2 marker

expression was comparable between WT and *Chil1*^{-/-} KCs. During MASLD progression, the pro-inflammatory signature in *Chil1*^{-/-} KCs was further enhanced, while anti-inflammatory marker expression became dysregulated (revised Figure S5C). Together, these data indicate that *Chil1* deficiency does not impair macrophage differentiation per se but biases KCs toward a partially pro-inflammatory, M1-like phenotype, providing additional context for the enhanced glycolytic flux observed in *Chil1*-deficient macrophages (Revised manuscript, page 7-8, line 307-321).

(7) The authors use the PDK activator PS48 and the ATP synthase inhibitor oligomycin to argue that increased glycolytic flux at the expense of OXPHOS promotes KC death. However, given the high energy demands of KCs and the fact that OXPHOS yields 15-16 times more ATP per glucose molecule than glycolysis, the increased apoptosis observed in Figure 4C-F could primarily reflect energy deprivation rather than a glycolysis-specific mechanism.

We thank the reviewer for highlighting this important point. We agree that KCs are highly metabolically active and that perturbations of OXPHOS can influence overall cellular energy balance. As noted in our response to comment #3, we further performed glycolysis inhibition assay by 2-DG *in vivo*, the protection of KCs observed following 2-DG *in vivo* (Revised Figure 4E-G) further provides evidence that increased glycolytic flux is not merely correlated with, but functionally contributes to KCs loss in

MASLD.

(8) In Figure 1C, KC numbers are significantly reduced after 4 and 16 weeks of HFHC diet in WT male mice, yet no comparable reduction is seen in Clec4Cre control mice, which should theoretically exhibit similar behavior under identical conditions.

We thank the reviewer for this comment. During our analysis, we indeed observed no reduction in KCs in the *Clec4f* cre control mice. This prompted us to consider that Cre insertion itself might influence KCs maintenance. To investigate this, we performed TIM4/Ki67 co-staining, which revealed significantly higher numbers of proliferating KCs in *Clec4f* cre mice compared with C57BL/6J mice under NCD. Following HFHC feeding, KCs proliferation in *Clec4f* cre mice increased even further. These results indicate that Cre insertion enhanced KCs self-renewal in *Clec4f* cre mice, which contributes to maintenance of the KCs pool during MASLD (Revised Figures S8A and S8B). (Revised manuscript, page 9, line 363-370).

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

To address the concerns raised in the public review, the authors should:

(1) Reassess their conclusions using the same panels in flow and microscopy, e.g., the combination of CLEC4F, TIM4, and IBA1. This will allow resKCs (CLEC4F+TIM4+IBA1+), moKCs (CLEC4F+TIM4-IBA1+), and moMFs (CLEC4F-TIM4-IBA1+) to be accurately defined and hence their viability and numbers correctly assessed.

We thank the reviewer for this insightful suggestion. In our flow cytometry analysis, we did not detect a CD45⁺CD11b^{low}F4/80^{hi}TIM4^{low} population, indicating that monocyte-derived KCs (moKCs) have not emerged in our model at this stage. To more accurately quantify resident KCs (resKCs) in the current study, we replaced CLEC4F with TIM4 staining and enumerated TIM4⁺ as well as TIM4⁺TUNEL⁺ cells. These data were highly consistent with CLEC4F⁺TUNEL⁺ cell counts, confirming that moKCs are not involved in KCs death during early MASLD (Revised Figure 1A,B,E,F).

(2) Investigate why the number of KCs in controls and MASLD are so distinct between Figures 1 and 6.

We appreciate the reviewer's suggestions. Like we explained above, Cre insertion promotes KCs self-renewal (Revised manuscript, Figure S8). This enhanced proliferative capacity likely accounts for the relative preservation of KCs numbers in Clec4f-Cre mice during HFHC feeding, explaining the apparent discrepancy with WT mice (Revised manuscript, Figure 6D-E).

(3) Normalise the *tunel*⁺ cells based on the number of KCs in PP vs PC regions.

After normalizing KCs death to KCs numbers in periportal (PP) versus pericentral (PC) regions, we found the proportion was significantly higher in PV regions compared to CV regions at 8 weeks of HFHC feeding. We have therefore revised our texts. (Revised manuscript, page 5, lines 194-201).

(4) Demonstrate the viability of KCs *in vitro* across conditions.

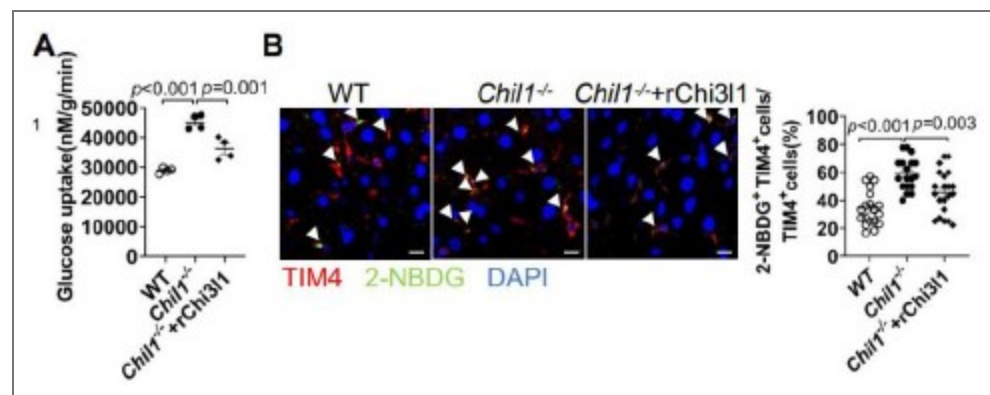
To confirm the identity and health of isolated KCs in our *in vitro* studies, we show that ~95% of primary isolated KCs are TIM4⁺ (Revised Figure S3A). Furthermore, Calcein-AM staining confirmed that the remaining KCs under our experimental conditions are viable and healthy (Revised Figure S4A).

(5) Confirm previous studies demonstrating different degrees of KC loss depending on the model of MASLD.

We thank the reviewer for highlighting this point. Consistent with previous studies, KCs loss has been reported to varying degrees depending on the MASLD model used, reflecting the heterogeneity of hepatic macrophages, marker choice, mouse husbandry, and diet regimen. For example, in a 6-week MCD feeding model, ~10% of CLEC4F⁺ KCs were TUNEL⁺ (Figure 4A, Ref. 9). Another 6-week MCD study reported a drop from 66% to 26% TIM4⁺ KCs (Figure 2A, Ref. 12). In an HFD model, TIM4⁺ KCs decreased by ~20% after 16 weeks (Figure 1G, Ref. 1). In a Western diet model, TIM4⁺ KCs decreased by >50% at 36 weeks (Figures 1J and 2C, Ref. 2). Together, these studies underscore the model-dependent nature of KCs loss and highlight the importance of experimental context and marker selection when assessing KCs dynamics in MASLD. We have included these studies in our discussion section (Revised manuscript, page 9-10, line 393-402)

(6) Demonstrate *in vivo* that loss of CHIL1 drives further glycolysis in KCs.

In Figure 6G-H of our previous study, we showed that Chi3l1 deficiency leads to more glucose uptake by KCs *in vivo* whereas supplementing KO mice with recombinant Chi3l1 will significantly reduced glucose uptake by KCs through treating mice with a fluorescent glucose analog 2-NBDG. We included the related figure here as Author response image 3.



Author response image 3. Chi3l1 limits glucose uptake by Kupffer cells *in vivo*. (A) Measurement of 2-NBDG (a fluorescent glucose analog) uptake by KCs *in vivo*. WT and *Chi1*^{-/-} mice, either untreated or supplemented with rChi3l1, were injected intraperitoneally with 12 mg/kg 2-NBDG. After 45mins, KCs were isolated and glucose uptake assessed by spectrophotometry. (B) Representative immunofluorescence images of liver sections stained for TIM4 (red) and 2-NBDG uptake (green) to visualize glucose uptake by KCs *in situ*. Scale bar = 10 μm (zoom). Quantification is shown as the percentage of TIM4⁺ cells that are also 2-NBDG⁺. Representative images were shown in B. One-way ANOVA was performed in A, B. P value is as indicated.

(7) There is no mention of the publication of the metabolomics dataset; this should be released with the manuscript.

We included the raw metabolomics dataset as Table S1 and S2 now.

Reviewer #3 (Recommendations for the authors):

(1) Methods: Reconsider which methods are described in the main text versus the Supplementary Information to improve readability and consistency.

Thank you for your valuable suggestion. We have reevaluated and adjusted the placement of the methods section between the main text and the supplementary materials.

(2) Line 34: Check for grammar issues.

L34 has been revised as follows : Additionally, using *Chi3l1*-deficient mice, we further demonstrated that increased glucose utilization accelerates KCs death *in vivo*.

(3) Lines 101, 110: Explicitly reference the corresponding Supplementary Methods sections.

We have included the references for these two methods sections (Revised supplementary materials and methods, Line 30, 65, respectively).

(4) Figure 2: *Iba1* marks all macrophages, not only monocyte-derived macrophages; both figure and text (line 205) require correction.

We have corrected *Iba1* represent hepatic macrophages including both KCs and MoMFs (Revised Figure 2C, manuscript page 5, line 182).

(5) Line 218-219: Avoid overinterpretation, as only KCs, hepatocytes, and hepatic stellate cells were assessed - not all hepatic populations.

We appreciate the reviewer's valuable suggestion and rephrased our description accordingly (Revised manuscript, page 5, line 186-189).

(6) Line 262: Use abbreviations consistently throughout the manuscript.

We have gone through the whole manuscript and double checked the abbreviations.

| (7) Line 264: Include the palmitic acid (PA) concentration used.

We included 800 μ M PA in the revised manuscript (Revised manuscript, page 6, line 250)."

| (8) Lines 316-317: Check for grammar errors.

Grammar errors are checked (Revised manuscript, page 8, line 340-341).

| (9) Line 337-338: See comment above on gating strategy.

We updated gating strategy accordingly (Revised manuscript, page 9, line 361-362).

| (10) Line 343-344: Note that *Chi3l1* is not exclusively expressed by KCs.

We rephrased our words accordingly (Revised manuscript, page 9, line 374-378).

| (11) Lines 355-358: The statement that "sustained glycolytic hyperactivation culminates not in sustained activation, but in apoptotic cell death" is unsupported by data or literature, as macrophage polarization was not analyzed in this study.

We removed the statement from the revised manuscript.

| (12) Lines 375-379: Rephrase to clarify that while KCs are metabolically active and glucose-demanding, excessive glycolytic flux accelerates apoptosis.

We have rephrased to clarify (Revised Manuscript, page 10, lines 405-407).

| (13) Lines 375-385 & 387-397: Consolidate overlapping statements for conciseness and coherence.

We have consolidate the overlapping statements (Revised manuscript, page 10, lines 405-425).

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